Recent Development of Deep Learning based EEG Network Analysis for Mental Health Disorders: A review

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Abstract

This paper reviews the recent advancements in Deep Learning-based Electroencephalography (EEG) Network Analysis for diagnosing and understanding mental health disorders. EEG, a non-invasive tool providing high temporal resolution of cortical activity, yields rich signals ideal for network analysis. Mental disorders, often characterized by aberrant functional and effective brain connectivity, present complex challenges for conventional clinical methods, driving the need for sophisticated analytical tools. This review systematically covers the methodologies used to analyze EEG signals, starting with signal processing and feature extraction (e.g., time, frequency, and time-frequency domains), followed by an examination of both traditional Machine Learning (ML) classifiers (e.g., LDA, SVM) and Deep Learning (DL) architectures (e.g., MLP, CNN, RNN/LSTM, DBN, and GCN) used for signal classification. Crucially, the paper details the construction of functional connectivity (FC) and effective connectivity (EC) networks, focusing on classical measures such as coherence, phase synchronization, and Granger Causality. The application of these techniques is then comprehensively discussed across several major mental health disorders, including Autism Spectrum Disorder (ASD), Depression, Epilepsy, Schizophrenia (SZ), Anxiety Disorder, Alzheimer's Disease (AD), OCD, PTSD, and Internet Addiction. Key findings consistently point to altered network topology (e.g., changes in clustering coefficient, path length, and efficiency) and synchronization patterns (e.g., hypo-/hyper-connectivity across specific frequency bands) as biomarkers. Deep learning models, particularly hybrid CNN-LSTM and GCN architectures, demonstrate superior performance, often achieving high classification accuracy by automatically extracting and fusing spatio-temporal features, thus holding significant promise for objective and automated clinical diagnosis.

Keywords Deep Learning, EEG, Brain Network Analysis, Functional Connectivity, Effective Connectivity, Graph Theory, Mental Health Disorders, EEG Biomarkers, Neural Networks

1 Introduction

Brain has approximately 100 billion neurons, a kind of nerve cells that can generate signals when they are activated. [86] Recently, EEG signal is widely used in the cinical field. Electroencephalography (EEG) provides a direct measure of cortical activity with millisecond temporal resolution[63], and records brain waves. EEG signals provide rich information and brain biomarkers[107]. Through the analysis of these EEG signals, the routine clinical diagnosis and prognosis of mental disorders can be effectively solved.[63] Most of the time, EEG is performed through a noninvasive device that captures the electrical brain activity produced by its upper layers. It consists of an array of electrodes which are placed over the scalp of the patient. [116]. Mental disorder encompass a wide spectrum of neurological and psychiatric conditions, are increasingly linked to abnormal connectivity patterns and topological alterations within the brain's functional networks. This paper aims to outline and discuss the recent development of EEG Network Analysis for several common mental health disorders, including Autism, Anxiety disorder, Bipolar disorder, Eating disorder, Schizophrenia, Insomnia, Obsessive-compulsive disorder(OCD), Post-traumatic stress disorder(PTSD) and Internet addiction.

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2 Methods and Materials

A literature search was performed using Google Scholar, ScienceDirect, PubMed and other relevant research databases for papers published recently. The terms 'Autism', 'Anxiety disorder', 'Bipolar disorder' ,'Eating disorder', 'Schizophrenia', 'Obsessive-compulsive disorder', 'Post-traumatic stress disorder', 'Internet addiction' and 'mental disorder' were searched in conjunction with EEG. Other relevant articles were found through citations from papers in this search. Papers exploring other technologies in regard to EEG and mental disorder research were excluded as they fell outside the scope of this work. It should be noted that because of the variable use of keywords in different articles, some relevant papers may have inadvertently been overlooked.

3 EEG signal processing

3.1 Processing of EEG signals

Different components constitute the processing of EEG signals: 1) Signal acquisition unit, 2) Feature extraction unit, 3) The decision algorithm. [3] After these steps, the EEG signals will then be passing to the external devices and examined.

3.1.1 EEG signal acquisition

The signal acquisition is traditionally represented by invasive or non-invasive electrodes. In obtaining bio-signal, initial elements are electrodes, which convert bio-potential signals from bio-potential sources into electrical signals.[151] Thus, signals could be recorded through invasive or non-invasive technologies. [151, 171] There are vasious non-invasive methods, including ElectroEncephalography (EEG), functional Magnetic Resonance Imaging (fMRI), Near Infra-Red Spectroscopy (NIRS) and, Magneto Encephalography (MEG).[79] In this paper, we focus more on ElectroEncephalography (EEG), one of the most popular non-invasive methods among various non-invasive methods.

Non-invasive EEG electrodes are positioned on the scalp using a standardized placement scheme to record EEG signals[1, 129]. Two types of electrodes are used for recording EEG signals, i.e., 1) wet electrodes and 2) dry electrodes.[171] As for wet electrodes, EEG gel or paste must be used together between the electrode surface and the scalp for proper skin preparation.[151, 171] This will reduce the impedance value.[171] Usually, wet electrodes are often made of silver chloride (AgCl)[171] and using this method also requires that the recording time need to be limited for several hours. This is because drying the gel or paste in time, as well as the subject's perspiration and movements (eye blinks, muscle movements, heart beats, etc.), can easily affect the measurement performance negatively.[151] Another approach -using dry electrode- does not need additional gel or paste and does not have time limitation. [151, 171]

EEG is the most popular signal acquisition method because of the high temporal resolution, safety, and ease of use. [56] However, EEG includes low spatial resolution, poor signal-to-noise ratio, and it is non-stationary in nature.[58]

3.1.2 EEG signal analysis

The neural network is consist of various artificial neurons which could help in creating nonlinear boundaries for decision. [107]. Different with 2D signals, such as images and speech, EEG signals are much more complicated and truly challenging to design an appropriate network for EEG-based classification tasks. Advanced algorithms such as machine learning and deep learning are developed for both the features extraction from raw data and the signal classification[60]. Decision-making was performed in two stages: 1) A feature extraction strategy and 2)A signal classification using learning-based algorithm classifier.[63] The performance of the neural model was examined in terms of training performance and classification accuracies and the results confirmed that the proposed scheme has potential in classifying the EEG signals.

Feature extraction Feature extraction is a process whereby the information or characteristics from the EEG signal is extracted so that these features can be easily interpreted[9], and the most important ones can be used for classification exercise.[1]

The features of EEG signals can be examined in four dimains: [107]
1. Time-domain features (TDFs)
2. Frequency domain features (FDFs)
3. Time and frequency domain features (TFDFs).
4. Features based on statistics.

Signal classification Signal classification can be solved by various analysis methods using machine learning or deep learning.[1, 107] Usually, the EEG signals can be classified through different classifiers.[62] A classifier has different parameters that can be learned from training data.[62] Based on different classifiers, the classification methods for EEG signal can be divided into five classes: [62]
1. Linear classifier
2. Nonlinear Bayesian Classifiers
3. Nearest Neighbor Classifiers
4. Neural Networks (NN)
5. Fusion of classifiers

In this paper, we focus on the neural network (NN), which is the most popular methods for the classification of EEG signals.

3.1.3 The decision algorithm

The decision unit, in brain computer interface (BCI) for example, is a hybrid unit with the purpose of classification, decision-making, and then passing the decisions to external devices , outputting the intention of the subject.[166]

3.2 Machine Learning and Deep Leaning in feature extraction and signal classification

3.2.1 Machine Learning Analysis

Conventionally, regression is used to build a relationship between EEG features, including event-related potentials (ERPs), behavioral scores, and seeking to determine which feature in time or frequency domain is involved in the processing of cognitive or perceptual states by analyzing each sample separately. However, these approaches can only handle univariate cases, whereby machine leaning analysis provide a chance in decoding and characterizing task-related brain states and extract them from non-informative high-dimensional EEG data.[60]

EEG features Discriminating features is a fundamental step before implementing machine learning in EEG analysis.[60]. For those EEG data which are spontaneous, researchers tend to study the signal in the frequency domain by characterizing frequency rhythms and amplitude.[60] Various other features can also be used for machine learning analysis. For instance, features like spectrum has been used by previous studies to detect the phase coupling between different frequency signals.[37]. Coherence, describing similarity of the frequency components across EEG channels[135], and connectivity, which assess the similarity of the time series across EEG sensors[122], are also useful features. Moreover, in a recent study, entropy, describing the distribution of EEG signal, has also been used.[164]. For evoked potentials (EPs) and event-related potentials (ERPs), latency and magnitude of peaks in time domain and in time-frequency domain can be used as features.[60]. Several other features, like the joint time-frequency (the variation of frequency features in relation to time) shown in figure 1 [122], the power (e.g., power spectral density), the phase value (e.g, phase-locked value) of each time-frequency point could be features for evaluating data that are spontaneous EEG or EPs/ERPs.[60]

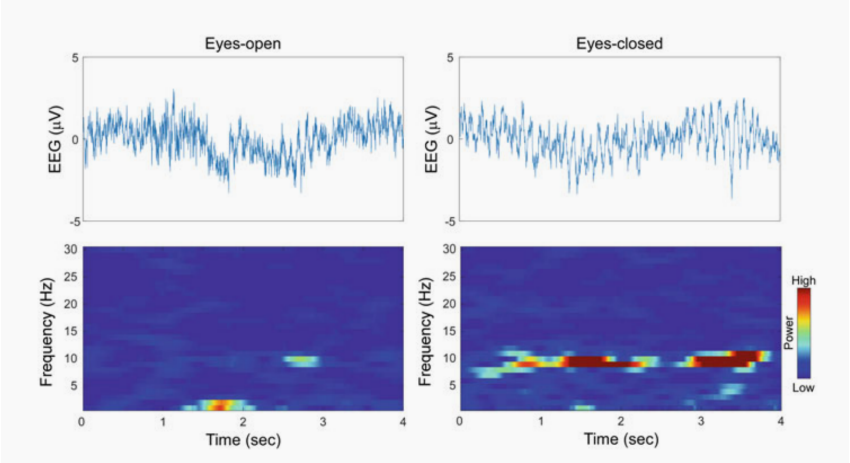


Figure 1: EEG time series (upper) and time-frequency representations (lower)[60]

Training Before training the classifier and learning the relationship between feature vectors and their class labels, it is necessary to split data multiple times into a training set (for training a classifier) and a testing set(for evaluating the performance of a classifier).[60] The cross-validation approaches, including "leave-one-out" cross-validation (LOOCV) and K-fold cross-validation have been introduced for organizing the training and testing samples.[59, 60]

Dimension Reduction or Feature selection For the purpose of reaching a higher classification accuracy, dimension reduction is of vital significance for recognizing features from high-dimensional EEG data.[60]. This process can be realized with or without labels, i.e. supervised or unsupervised methods. Principle component analysis (PCA) is a traditional approach for dimension reduction, which is based on the projection of high-dimensional data along a direction for which has the maximized variance of data in a linear subspace spanned by a small number of latent component, in other words, PCA uses orthogonal transformation. [67] Unlike unsupervised methods, which reduce the dimensional without the classes, supervised methods exploit class information and thus can map high-dimensional data into a low-dimensional space where different classes are well separated.[94] For supervised feature selection, there are several techniques can be used to discriminate different classes. Univaraite statistical techniques are used, including t-test, ANOVA, and Pearson correlation coefficient. For multidimensional EEG data, there are wrapper techniques, like recursive feature elimination (RFE), least absolute shrinkage and selection operator (LASSO) and elastic net approach.[60]

Classifier After the feature extraction using the most informative features, the data needed to be forwarded to an appropriate classifier.[84] A classifier learn the function between features and classes which are categorized into discriminative models. The purpose of a discriminative model is to directly evaluate a prediction function with a given parameter.[60] In this section, we explain several prominent classifiers.

Linear Discriminant Analysis (LDA). The LDA constructs a predictive model and finds a projection direction to a line such that all data representing different classes are well separated.[60, 130] This model does not need high computational requirement. A simple example of a linear combination of the features could be:

$$X_w = x_1 w_1 + ... + x_k w_k \tag{1}$$

and the classifier separate class A and B based on the criteria $X_w > 0$ and $X_w < 0$ respectively. [60]

Support Vector Machine (SVM). Through building a hyperplane in a high-dimensional space, SVM facilitates the process of achieving an optimal decision boundary, which maximize the margin between boundaries of different classes. This technique can be applied in both linear and nonlinear cases. For non-linear version, the optimal combination can be defined using a grid-search via inner cross-validation.[60] Compared to the original SVM, the LS-SVM has lower computational cost since it use linear equations instead of quadratic and constrain via equality instead if inequality. [142]

Naive Bayes classifier (NB). In contrast to SVMs, NBs learn statistical model that a sample can be generated from a particular class and then predict.[61] A NB is based on the Bayes theorem with the

assumption that all features are independent of each other for a given class. Typically, it is also assumed that is the feature variables are continuous, then they have a Gaussian distribution and is called GNB classifier[60]. Additionally, using the chain rule, the results of this classification method is decided by the posterior probability $P(y|x_1, x_2, x_3, ..., x_n)$. [161]

K-Nearest Neighbor algorithm (KNN). The KNN algorithm is a straightforward classification method based on observing and dividing the training objects via the closest feature. Usually, an object is assigned to a class most common amongst is k (a small positive integer, which is the only parameter that can be adjusted in this model) nearest neighbours. For instance, $k = 1$ means the object is assigned to the class of that single nearest neighbour. Besides, the value of $P(y|x)$ is computed as the ratio of members of class y among the k nearest neighbours of x.[130]

3.2.2 Deep Learning Analysis

Brain-computer interface (BCI) technologies is a traditional tool of analyzing EEG signals via direct communication between humans and computers. However, due to the unsatisfactory recognition effects of EEG signals in the classification tasks, more advanced techniques like deep learning have been developed recently. [60] Deep learning is a particular subset of machine learning using artificial neural networks (ANNs), which are developed from the structure of neurons in human brain, which is consist of multiple processing layers.[60] Outperforming than the traditional machine learning algorithms, deep learning is a better technique to learn and understand complex representations directly from the raw signal. Additionally, it can automatically extract advanced features for classification and has been applied various fields.[60] Recent deep learning architectures are based on three classes of ANNs: Multi-Layer Perceptron (MLPs), Convolutional neural network(CNNs), and Recurrent Neural Networks (RNNs). MLPs learn a representation from all data input, CNNs exploits the spatial structure of data by computing filter-selective feature maps, while RNNs process sequential data introducing the concept of memory.[12] In the following part of this section, we will explain several deep learning algorithms.

Multi-Layer Perceptron Network (MLP) Multi-Layer perceptron (MLP) is an ANN technique which tend to simulate the way that how neurons work in the brain.[169] It represents a generalization of the single layer perceptron. While A single layer perceptron forms half -plane decision regions, MLP can form arbitrarily complex decision regions and can separatevarious input patterns.[7]
An ANN adopting MLP is composed of three layers, as figure.2 shows. The first layer is the input layer with nodes representing the number of features in the training set. The second one may be constituted by numerous hidden layers, and the number of hidden layers and the containing unites are arbitrary. The final layer is the output layer.[169] Between the nodes, the connecting lines carry some weights that represent the amount of electrical impulses in the neurons and are usually learned and updated during the training via back-propagation learning. At each nodes in the hidden layers, sum of the products from each input node and the corresponding weight is calculated as the input for a transfer function (activation function), simulating the excitation/inhibition of the electrical impulses. Some common transfer functions used in MLP are step function, Linear function, and Sigmoid function. [169]

Convolutional neural network (CNN) CNN is a type of deep and feed-forward ANN based on the connectivity of neurons in the animal visual cortex. CNNs can assess nonlinear features via convolutions and nonlinearities and through multiple layers of processing, CNNs can represent higher-level features as compositions of low-level features. Since CNN has good performance in decoding and iterative training, it has been successfully in EEG analysis. In EEG studies, CNN enable features to be extracted from raw data via two convolutional layers to evaluate the input signals, such as time points and electrodes. Conventionally, CNN is composed by an input and an output layer with multiple hidden layers.[60] Here, a convolution kernel is applied to the input data to produce a feature map and these CNNs can learn spatial features from the input data.[116] Meanwhile, deep learning can be combined with various existing analysis methods, one example is the CNN. Through the modification of the filter-bank common spatial patterns methods, EEG signals are converted into new temporal representations and a CNN can be utilized foe motor imagery (MI) EEG data classification. [118]

Recurrent Neural Networks (RNN) A RNN is an extended feed-forward neural network, dealing with the variable-length sequence input by having a recurrent hidden state whose activation at each time is dependent on that of the previous time. One apparent limitation of common RNNs is that it is hard

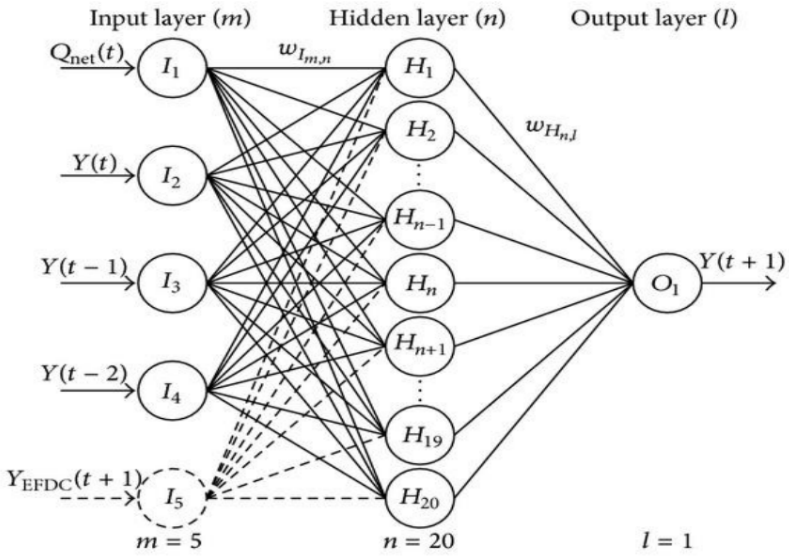


Figure 2: A typical Multi-Layer perceptron network [7]

to train since the vanishing/ exploding gradient problems. [2, 60] Therefore, long short-term memory (LSTM) has been developed as the gated recurrent network architectures to optimize those standard RNNs. [39]

Long short-term networks (LSTM). The most commonly used LSTM architecture, which is shown in figure3, is circulatory adn has been described as follows: [2, 60]
Forget gate, which is the first gate, is to determine the information that need to be discard from the cell state and the decision is made by a sigmoid layer:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{2}$$

The second gate is an input gate with a sigmoid layer(with updated values) and a tanh layer(which creates a vector using the updated values in equation 3 and 4)

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{3}$$

$$\tilde{C}_t = tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \tag{4}$$

Then we have the updated cell state C_t

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \tag{5}$$

Based on the updated cell state and a sigmoid layer, the final outputcan be calculated:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{6}$$

$$h_t = o_t * tanh(C_t) \tag{7}$$

Here, σ is sigmoid/logistic activation function which maps the resulting values in between 0 and 1,

$$\sigma(x) = \frac{1}{1 + e^{-x}} \tag{8}$$

tanh is hyperbolic tangent activation function which maps the resulting values in between -1 and 1,

$$tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \tag{9}$$

x_t is the input vector, $h_{t-1}$is the past hidden state, W_f, W_i, W_c, W_o are weight matrices, and b_f, b_i, b_c, b_o are bias vectors.

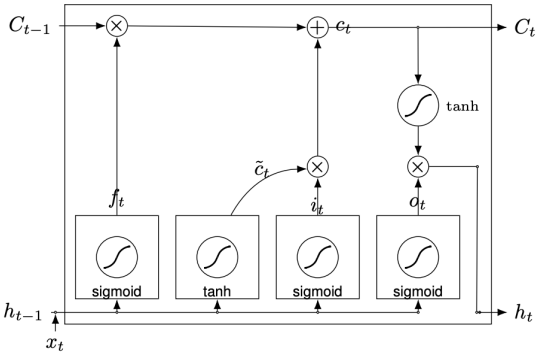


Figure 3: LSTM cell architecture [2]

Deep belief network (DBN) A DBN is a classical generative probabilistic model consist of some restricted Boltzmann machines (RBMs), in which the higher layer of DBNs tend to learn more complex feature representation. [60, 114] One of the most significant applications of DBN in EEG analysis is the dimension reduction. The reason is that through the visible layer, the high-dimensional data can be fed into the hidden layer, which can identify the characteristics of data based on the weights of connections between layers. Additionally, When training a DBN, it is best to train one layer at a time, in a layer-wise manner and once every layer of the model has been trained, the output of the final RBM can be used as the input for the classification. [60]

4 Network construction

Brain networks can be divided into structural(or neuroanatomical) and functional networks, which are derived from anatomical or physiological observations respectively.[18, 134] Structural connectivity depicts anatomical connections linking a set of neural elements, which can be revealed by the techniques such as magnetic resonance imaging (MRI) and diffusion tensor imaging (DTI).[18] Generally, functional connectivity (FC) is derived from time series observations, describing patterns of statistical dependence among neural elements. To obtain time series data, various techniques can be used, including electroencephalography (EEG), magnetoencephalography (MEG) and functional magnetic resonance imaging (fMRI). Different methods can be used to compute time series data, including cross-correlation, mutual information and spectral coherence. FC is highly time-dependent and exhibits non-stationary fluctuations. Effective connectivity (EC) captures a network which is of directed casual effects between neural elements.[134] In this paper, we focus more on the construction of EEG-based functional networks and effective networks and their application.

4.1 Connectivity analysis

There are various methods in detecting EEG connectivity, and they could be divided into effective connectivity(EC) and functional connectivity(FC) .[60] The measures in EC assess the casual relationship between EEG signals, while that in FC assess the linear or nonlinear statistical dependence between these signals, without giving any information about their casual relationship.[60] In order to define their statistical dependence, four types of measures could be used: coherence-based measures, phase synchronization-based measures, generalized synchronization-based measures, and granger causality-based measures.[60] In this section, we concentrate on introducing and discussing these four classical functional connectivity measures.

A. Coherence-based measures The coherence could assess the linear relationship between two signals at each frequency bin evaluated[60] and many studies have used it to quantify oscillatory interdependencies between brain areas.[122] The coherence between a pair of EEG signals can be dependent on the frequency band they are in, any pair may be coherent in some frequency bands and incoherent in others.[135]

a. The magnitude-squared coherence One of the coherence-based measures has been proposed by previous researchers.[98] After the transmission of EEG signals through the electrodes, fast Fourier transform (FFT) or discrete/continuous wavelet transform (DWT/CWT) can be used to convert the two time-depended signals X and Y into frequency domain, i.e. $X(t)$ and $Y(t)$ at each time point t. Then, we estimate the individual spectral power density function $S_{xx}(f)$ and $S_{yy}(f)$, and the cross spectral power density function $S_{xy}(f)$ for a specific frequency band f. Using these density, the coherency function $C_{xy}(f)$ can be calculated through the formula

$$C_{xy}(f) = \frac{S_{xy}(f)}{\sqrt{S_{xx}(f)S_{yy}(f)}} \tag{10}$$

Using $C_{xy}(f)$ we can get the coherence at each frequency band f

$$COH_{xy}(f) = |C_{xy}(f)^2| \tag{11}$$

Based on the knowledge of statistics, the coherence value $COH_{xy}(f)$ is ranging from 0 to 1. Larger value suggests higher linear dependence between two EEG signals, while 0 means that there is no linear relationship. Therefore, apparently, the coherence-based measures can only be applied when estimating the linear relationship while it cannot detect the non-linear relationship. Another limitation of this method is that it fail to separate the amplitude and phase of EEG signals.[60] Additionally, since this method is not able to provide casual information about EEG signals, this is regarded as FC measure.

b. Imaginary coherence (ImC) For the purpose of reducing the effect of the volume conduction[52], the imaginary coherence (ImC) has been introduced and it is defined as[99]

$$ImC(f) = \frac{|S_{xy}(f)|}{\sqrt{|S_{xx}(f)||S_{yy}(f)|}} \tag{12}$$

c. Phase Slope Index (PSI) In order to detect some casual dependence, the Phase Slope Index (PSI) is developed by some researchers, which is an EC measure.[137] Since the interactions between neural processes require some time, suppose the travel speed at different waves are almost the same, then the phase difference between the sender and recipient has a positive correlation with frequency. The researchers also assumed that the slope of the phase spectrum is positive. Based on these assumptions, the PSI is defined as follows ——formula here——

B. Phase synchronization-based measures The phase synchronization-based measures are useful in exploring the nonlinear dynamics in chaotic systems of EEG signals, but fail top assess the casual relationship.[60] However, unlike the coherence-based measures, the phase synchronization-based measures could separate the phase component and amplitude component of EEG signals for a given frequency or frequency band during assessing connectivity.[60]

a. Index: PLV Suppose the phase time series $\phi_x(t)$ and $\phi_y(t)$ of EEG signals $X(t)$ and $Y(t)$, and the $\Delta\phi_{xy}(t)$ is their phase difference. The principle of the phase synchronization (PS) measure is to detect the existence of a phase locking between two systems defined as

$$\Delta\phi_{xy}(t) = |\phi_x(t) - \phi_y(t)| \leq constant \tag{13}$$

If Eq.13 holds, then phase synchronization is detected. Then for practical analysis, we use the relative phase difference that is in the interval $[0, 2\pi)$ and thus usually the other formula is needed:

$$\Delta\phi_{rel}(t) = \Delta\phi_{xy}(t) mod 2\pi \tag{14}$$

The first step is extracting the instantaneous phase of each EEG signal using the Hilbert transform or the wavelet transform.[52] Then, in order to quantify the phase synchronization between EEG signals using the phase difference, the index Phase Locking Value (PLV) can be used.

$$PLV_{XY} = |\frac{1}{N}\sum_{n=1}^{N} e^{i\Delta\phi_{rel}(t_n)}| \tag{15}$$

Here N is the number of time points of $\Delta\phi_{rel}(t)$, which means that this is a value based on the average of time. The PLV is ranging from 0 to 1, while $PLV = 0$ means the phase difference is uniformly distributed over $[0, 2\pi)$ and $PLV = 1$ means n the phase difference time series is constant across the entire time series. Although this index will be highly sensitive to the common sources problem, it is widely used in traditional EEG analysis. [60]

b. Other indexes: PLI and WPLI Other advanced indexes, such as the phase lag index (PLI)[137] and the weighted phase lag index (WPLI)[157] are also utilised by researchers to assess the relationship between the EEG signals of two brain regions.

C. Generalized synchronization-based measures Other than the Phase synchronization-based measures, here is another type of synchronization, namely the Generalized Synchronization. The generalized synchronization-based measures could provide a normalized estimation of dynamical interaction between multi-electrode EEG signals, while the Phase synchronization-based measures can only detect that between two signals.[60] The generalized synchronization exists when the state of the response dynamic system Y is a function of the state of the driving dynamic system X.[138]

a. Synchronization likelihood (SL) Among various algorithms that can be used to detect the generalized synchronization between EEG signals, the mostly used one is the synchronization likelihood (SL).[60] This technique is proposed by[138]. Suppose M is the number of brain regains, $k = 1, ..., M$ is the index of these regions. N is the total number of time point of EEG signals, and $i = 1, ..., N$ is the index of the time point. Thus we have the EEG data is $x_{k,i}$. For multi-electrodes EEG signal, the vector $X_{k,i}$ is defined as

$$X_{k,i} = (x_{k,i}, x_{k,i+\tau}, x_{k,i+2\tau}, ..., x_{k,i+(d-1)\tau}) \tag{16}$$

Here τ is the time lag and d is the embedding dimension.
For each brain region (or electrode) k and each time i, the probability $P^{\epsilon}_{k,i}$ of that embedded vectors are closer to each other than a distance ϵ is defined as:

$$P^{\epsilon}_{k,i} = \frac{1}{2(w_2 - w_1)} \sum_{j=1, w_1 < |i-j| < w_2}^{N} \theta(\epsilon - |X_{k,i} - X_{k,j}|) \tag{17}$$

where θ is the Heaviside step function ($\theta(x) = 1$ when $x \leq 0$, otherwise θ), $w1$ is the Theiler window which is used to avoid the auto-correlation effects and should be at least of the order of auto-correlation time, and w2 is a window that sharpens the time resolution of the synchronization measure and is chosen such that $w1 \ll w2 \ll N$.
Now for each specific k and i, a critical distance $\epsilon_{k,i}$ is determined by $P^{\epsilon_{k,i}}_{k,i} = P_{ref}$, where $P_{ref} \ll 1$. For each pair of time point (i, j) satisfying that $w_1 < |i - j| < w_2$, the number of channels $H_{i,j}$ where the distance between $X_{k,i}$ and $X_{k,j}$:

$$H_{i,j} = \sum_{k=1}^{M} \theta(\epsilon_{k,i} - |X_{k,i} - X_{k,j}|) \tag{18}$$

And through calculation we can know that $H_{i,j}$ is smaller than the critical distance $\epsilon_{k,i}$. Then the criteria for evaluating the synchronization likelihood $SL_{k,i,j}$ is that if $|X_{k,i} - X_{k,j}| < \epsilon_{k,i}$ then

$$SL_{k,i,j} = \frac{H_{i,j} - 1}{M - 1} \tag{19}$$

Otherwise

$$SL_{k,i,j} = 0 \tag{20}$$

Then use $SL_{k,i}$ to assess how strongly electrode k at time i is synchronized to all the other $M - 1$ electrodes, we average over all the j

$$SL_{k,i} = \frac{1}{2(w_2 - w_1)} \sum_{j=1, w_1 < |i-j| < w_2}^{N} SL_{k,i,j} \tag{21}$$

SL value is ranging between P_{ref} and 1. If $SL = P_{ref}$, then all the M electrodes are unrelated. If $SL = 1$, then a maximum synchronization between these M electrodes is revealed.

D. Granger causality-based measures The Granger causality (GC) can assess the casual relationship or direction of information flow between different neural oscillations, except the PSI. It is one of the prototypical data-driven effective connectivity (EC) techniques.[119] The basic idea of GC measures is that if time series $X(t)$ is the GC of another time series $Y(t)$, then predictions of the further value of $Y(t)$ based on its own past values and the past value of $X(t)$ are better than predictions of the further value of $Y(t)$ only based on its own past values. And below we introduce some mathematics statements that has been developed to illustrate GC.

a. VAR models The conventional GC measures are based on the linear stochastic auto-regressive (AR) models,[45] and Granger defined the causality between time series based on the following two assumptions: (1) the cause must happen prior to its effect; (2) the cause has information that could improve the prediction of the effect, above and beyond the information contained in the past of the effect. [60] A simple condition of using this measure is to test the relationship between pairs of processes, using the AR model. For multivariate Granger causality analysis, a vector auto-regressive (VAR) model is used. The VAR scheme involves two main steps: 1) model estimation and 2) model inference to estimate brain connectivity.[87]

b. State-space models The state-space models are extensive models considering a more general class of linear stochastic processes in the form of state-space equations to model EEG dynamics. [87]

4.2 Graphs

4.2.1 Blocks of networks

A graph is a mathematical representation of a network, consisting of a set of nodes which are connected by the edges[31, 155]. The values (for instance: strength, length or importance of the connection) of the edges can be presented in an adjacency matrix, which represent the structural or functional relationship between nodes[155]. Therefore, this matrix contains all information of the network and extensive information on graphs.

4.2.2 Forms of functional networks

Functional networks can be divided into different types. By setting a threshold for the functional connectivity above where a functional connection is present, a binary network can be obtained. Such a binary network can be either weighted or unweighted. While a weighted network takes into account of value of the edges between nodes, in an unweighted network, only the existence or absence of an intersection matters and the edges either exist or not[17, 155]. Additionally, a network can be directed or undirected. Utilizing an effective connectivity measure, which can infer the information on the directionality of communication, a directed network can be built. However, undirected networks are more commonly used nowadays although the potential advantage of providing useful additional information of the directed networks [156]. The brain can be characterized as an unweighted and undirected network, where EEG electrodes serve as nodes and a link between any pairs of nodes represents a correlation of time series or other statistical measure associated with these nodes. Through this way, the topological properties of functional networks based on EEG signals can be analyzed[68].

4.2.3 Network metrics

The most frequently used measures are clustering coefficient C, the path length L, degree k, degree distribution $P(k)$, degree correlation, moduarity, centrality and graph spectral analysis[155].In this section, we explain several commonly used measures.

Clustering coefficient The clustering coefficient is a measure of local connectivity, and is defined as to what extent the neighboring nodes to which a node is connected are connected with each other[155]. Thus, if node A is connected to node B, and B is connected to C, A tends to connect to C. In order to quantify this phenomenon, a clustering coefficient was introduced. A combination of three nodes with (at least) two edges is called a triplet, and a closed triplet is defined when all the nodes are connected.

The average clustering coefficient C, which represents the functional segregation ability of the brain, is computed by dividing the number of closed triplets by the total number of triplets[68, 172]

$$C = \frac{1}{N} \frac{\sum_{i,j} a_{ij} a_{ik} a_{jk}}{k_k(k_k - 1)}$$

Here N is network size, a_{ij} is the corresponding element of the adjacency matrix between nodes i and j, and k_i is degree of node i that is obtained by summing all coming links to i.

Shortest path length The shortest path length is the smallest number of edges between two nodes, which represent the size of the network.The average shortest path length L is thought as a measure of global efficiency. It is defined as the mean number of steps along the shortest paths between all possible pairs of network nodes.[155, 172]

$$L = \frac{1}{n} \sum_{i \in N} L_i$$

Here n is the total number of nodes in set N, L_i is the mean shortest path length of node i, and it can be described as

$$L_i = \frac{1}{n-1} \sum_{i \neq j \in N} I_{ij}$$

where I_{ij} is the shortest path length between nodes i and j.

The degree The degree k of a node is the number of edges connecting this node to others, and it is defined as[172]

$$k_i = \sum_{j \in N, i \neq j} a_{ij}$$

The relative importance of nodes in a network can be revealed by the degree: nodes with high degree can be regarded as important points. Additionally, the degree distribution $P(k)$ reflects the probability that a randomly chosen node will have degree k. The measure 'degree correlation' indicates the extent that nodes with the same or different degree are connected to each other. Assortative networks are those with nodes showing a tendency to be connected to nodes with the same degree, while disassortative networks are those with nodes to be connected to nodes with a different degree. The EEG-based brain network seems to be assortative. [155]

The network efficiency The functional integration ability of the brain refers to the ability of extent of the brain network to combine the information of various brain regions, and it is represented by the global efficiency, which is defined as [68, 172]

$$E_{global} = \frac{1}{n(n-1)} \sum_{i \neq j \in N} \frac{1}{I_{ij}}$$

The local efficiency of node i is defined as[68, 172]

$$E_{i-local} = \frac{1}{n_i(n_i - 1)} \sum_{i \neq j \in N_i} \frac{1}{I_{ij}}$$

where N_i represents the collection of nodes which are directly connected with node i, n_i is the total number of nodes in set N_i. Moreover, the mean local efficiency, representing the functional integration ability of N_i when the node i is removed, can be calculated using [172]

$$E_{local} = \frac{1}{n} \sum_{i \in N} E_{i-local}$$

4.2.4 Network models

Various network models help putting network metrics into perspective, and several models have been widely used in the functional brain network studies. Using C, L and $P(k)$, four types of networks can be characterized: the random graph, the regular graph, the small-world network, and the scale-free network[155].

4.3 EEG based functional networks

The construction of brain functional networks Fairly different from connectivity measures, which only reveal how pairs of different brain regions are functionally connected, network analysis provides information on the organization of networks [136]. The construction of functional networks is based on the strength or consistency of functional interactions between the network nodes[153]. To construct functional or structural brain networks, signals from electroencephalography (EEG), magnetocephalography (MEG), or magnetic resonance imaging (MRI) are often used.[68] In this paper, we focus on constructing the functional networks using EEG signals.

To initiating network analysis, we should first define its nodes and edges in two separate spaces- sensor space and the source space. EEG network can be constructed within either or both of them. The sensor space is where multi-channel EEG record signals and the source space is where the brain activity actually occurs[180]. Regardless of which of the spaces are used for network construction, the steps are similar in these approaches.[80] For EEG-based brain functional networks, the individual or groups of sensors are utilized as nodes. Computing a measure of the associations between pairs of nodes is the first step, These measures include cross-correlation and coherence, synchronization likelihood, and S-estimator. The most frequently used metric for evaluating these associations is the correlation coefficient. Next, a threshold is applied, and all entries (i.e. all correlations) are compared with the threshold, thus the edges are set and are used to connect nodes. Then the matrix is reduced to a binary adjacency matrix (i.e. an unweighted graph). Finally, graph theoretical metrics are used to analyze the binary network, including the clustering coefficient, modularity, efficiency and etc[68, 180].

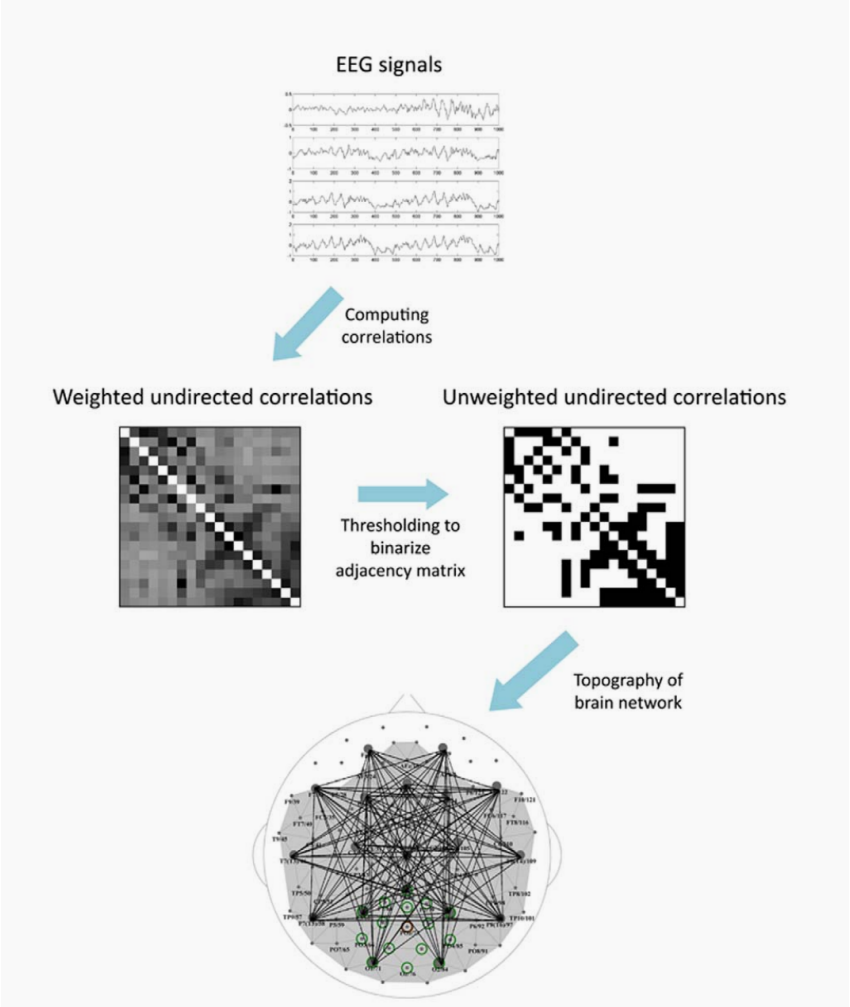


Figure 4: Construction of EEG-based brain networks[68]

4.4 Resting-state EEG

In clinical practice, EEG recordings are often used as a tool to help clinicians with diagnoses and prognoses [33]. Quantitative analysis of EEG signals from resting-state subjects is a low-cost and potentially useful neurophysiological approach. EEG band power analysis, which is the measure using the estimation of the power density of selected EEG frequency bands [92]. The most common way to characterize the resting-state EEG signals is to breaking into different frequency bands. The clinically relevant frequency bands of the EEG range from 0.3 to 100 Hz. The main frequency bands are Delta (1 to 3 Hz), theta (4 to 7 Hz), alpha (8 to 12 Hz), beta (13 at 35 Hz), and gamma(>35 Hz) [162]. Delta waves dominate deep sleep and is fundamental for the detection of event-related slow waves from attention and salience [74]. Theta is associated with memory processes. Alpha is typical of mentally inactive state with closed eyes and psycho-sensory rest [112]. Beta waves present information relating to alertness, active task engagement, and motor behavior. Additionally, gamma waves present during working-memory matching and a variety of early sensory responses [144].
Multiple analysis can be used in the diagnosis of mental disorder using EEG. For instance, spectrum powers can demonstrated the ability to discriminate different brain state of patients [44]. Entropy theory has also been applied in the time domain of EEG [167]. Functional connectivity is a crucial method for examining consciousness [120]. Spectral entropy analysis and some other connectivity approaches have been employed as well, such as weighted symbolic mutual information (wSMI), cross-approximate entropy, debiased weighted phase lag index (dwPLI), symbolic transfer entropy, and multivariate Granger causality[10].

4.5 Task-related EEG

When collecting task-related EEG data, we employ a specific task wherein subjects are presented with certain stimuli and/or asked to make responses. After being collected, the raw data are subject to preprocessing. The preprocessing procedure for task-related EEG is virtually the same as that for resting-state EEG, and rare differneces include epoch extraction, baseline correction, and bad epoch identification. Some common analysis of task-related EEG include time domain and time-frequency analysis. In order to extract more information from the raw data, more advanced techniques like source analysis and single-trial analysis are needed [177].

5 Application on different mental health disorders

5.1 Autism spectrum disorder (ASD)

5.1.1 Clinical backgrounds of ASD

Autism spectrum disorder (ASD) is characterized by changes of neurological functions, which will lead to typical consequent deficits in socialization, communication, and the presence of restricted and repetitive behaviors [112]. These symptoms are present from early childhood and are impairing to patients' everyday functioning [16]. The main diagnostic criteria comprises two symptom domains: 1) social/communication deficits (SCD), and 2) Fixated interests or restrictive/repetitive behaviors (RRB) [148]. Since the high prevalence of ASD, many attempts have been made to discriminate and detect ASD by analysing EEG or similar bio-signals.

5.1.2 Functional brain networks in Autism

The functional brain networks can be measured by multiple techniques, however, despite the increasing number of functional connectivity studies on ASD, little consensus on atypical functional network patterns has emerged [88]. In this section, we briefly reviewed several recent studies, which used resting-state and task-related EEG combining with other techniques to reconcile the discrepancies.

Resting-state EEG FC findings **Resting-state coherence.** ASD is characterized by a general long-range underconnectivity and sometimes a short-range (or local) overconnectivity and this has been recognized as a consensus by a majority of previous studies [27, 70, 88, 106], though EEG findings relevant to this hypothesis are still highly inconsistent [88]. For instance, coherence is a popular measure to assess the brain connectivity, but the EEG results pertaining to the local level are still not completely consistent,

with both local hypo- and hyper-connectivity measured by the coherence analyses due to different task designs [64, 88]. Coherence is positively correlated with the brain connectivity, which means that high coherence values are taken as a measure of strong brain connectivity [16]. Prior studies have demonstrated that resting state EEG coherence reduced in individuals with ASD [21, 89, 93]. Dickinson et al. [27] has quantified functional connectivity in the alpha band in ASD subjects to see whether group level features of circuit dysfunction are consistently present across the range of cognitive function seen in the autism spectrum. In this study, long-range hypoconnectivity was presented in children with ASD compared to TD children, with temporal interhemispheric connectivity showing the largest difference between the two groups, which reflecting white matter disturbances or increased signal variability at temporal sites in ASD [27].

Graph theoretical measures used in resting-state EEG. Graph theory has been utilized in investigating abnormal connectivity patterns and topological properties of ASD patients by using resting-state EEG data. For instance, Han et al. [49] aimed to discover the abnormalities of children with ASD from early childhood to late childhood by using resting-state EEG, applying clustering coefficient (CC) and the shortest path length (PL) to quantify the extent of local inter-connectivity in ASD children. From the graph analysis, a reduced CC and enhanced PL in specific frequencies was observed in children with ASD [49]. This coincides with another important recent study, whose results showed that in the theta band network integration measured using characteristic path length (CPL) increased as the cognitive load during the emotion regulation task increased, with a complementary trend in the CC [168]. Additionally, a more recent study, which applied the source-level resting state EEG and frequency-domain partial canonical coherence (PCC) method, to investigate whether there is any change has any correlation with age [78]. The results revealed an age-independent functional connectivity pattern in the theta band, consisting of the brain areas responsible for emotions and consciousness, where the strength of connections is higher in neurotypical children compared to ASD children [78]. Lower network global clustering in the delta and theta band in ASD children has also been found by Kurkin et al. [78], thus, more segregated, but more highly connected subnets are formed in the delta and theta band in neurotypical individuals compared to ASD ones. This suggested that neurotypical children have an increased control over emotions and stronger interaction between the emotional and conscious domains in [78]. Moreover, in the extended alpha band, the study showed an age-dependent functional connectivity pattern, demonstrating hyper-activation in the ASD group for ages below 6–7 years old and hypo-activation—for older ages [78].

Synchronization. Temporal synchronization analysis combined with the graph theory (including weighted, binary graph, and minimum spanning tree (MST)) was applied to capture the aberrant brain connectivity in ASD by Zeng et al. [175]. Whole brain connectivity in ASD group was found to be significantly reduced in theta and alpha band compared to TD [175]. Coinciding with precious studies, decreased PL together and decreased CC in ASD in alpha band indicated a loss of small-world architecture to a random network [175]. Besides, in MST analysis, ASD children showed a significant lower leaf fractions with a decrease tree hierarchy in the alpha band, indicating a shift towards line-like decentralized organization in ASD [175]. As for global Synchronization, Han et al. [48] analyzed multichannel resting-state EEG based on Rényi Entropy and found there is a significant increase of global synchronization in EEG of ASD children, which confirms that there are abnormalities in brain synchronization. In addition, EEG can capture oscillatory electrical activity, which is a phenomenon molded by inhibitory activity and can seen as part of the synchronization of neuronal [36].

Task-related EEG FC findings **Task-related coherence.** As for the coherence, recent studies have utilized different tasks into EEG recordings process, thus both increased and decreased coherence appeared in their results. For instance, EEG coherence during joint attention perception task is reduced in ASD [64]. Specifically, across conditions, adolescents with ASD showed reduced right hemisphere temporal–central alpha coherence compared to TD, revealing that greater right temporal–central alpha coherence during joint attention was positively associated with social cognitive performance in TD but not in ASD [64]. Besides, Buckley, Ashura W., et al. [16] investigated brain connectivity using EEG data from a large cohort of children that had typical development (TD), developmental delay (DD), or ASD during the awake, slow wave sleep (SWS), and rapid eye movement (REM) sleep state. Among the results from the measures used for detecting the brain connectivity (coherence, phase lag, spectral power, Pearson and partial correlations) in this study, the most striking was the increased coherence in ASD relative to TD, almost exclusively during SWS, concentrated in the frontal–parietal pairs [16]. While spectral power and Pearson and partial correlations showed almost no differences between ASD and TD in this study, decreased EEG phase lag was often associated with high coherence [16]. This

indicate that in ASD children there are not only highly coherent networks with an unvarying relationship among phases in different brain regions, but also a network with a remarkably high and rigid coupling of phase across brain regions [16]. The inconsistent results about hypo- or hyper connectivity of patients with ASD brain network may be influenced by the age, one study conducted by Haghighat, Hossein, et al. [47] based on the resting-state fMRI may support this statement. In this study, evaluation of the within-connectivity showed hyper-connectivity, especially in cerebellum and brainstem in ASD children but both hyper/hypo connectivity in adolescents and adults [47]. Moreover, in ASD children, difference in the between-connectivity among default mode network (DMN), salience-executive network and fronto-parietal network were observed. While there was a negative correlation between DMN and temporal network, Full correlation comparison between ASD adolescents and TD individuals showed significant differences between cerebellum and DMN [47]. The conclusion of this study is that the hyper-connectivity may appear in childhood, but both hypo and hyper-connectivity will appear after childhood and that abnormal resting connections in ASD exist in the regions of the brain known to be involved in social cognition [47].

Graph theoretical measures used in task-related EEG. Graph theory has also been widely used in the analysis of task-related EEG in detecting ASD. A comprehensive study in 2021, conducted by Wadhera et al. [159] has investigated the topological configurations of whole-brain functional network at local and global level under cognitive load in context of risk-perception task. The findings revealed that a wide portion of whole brain network is engaged and gets reorganized in ASD while perceiving the social cues (risk/safety cues). In both risk and safe situations, ASD patients showed more connectivity in the left hemisphere reflected by the higher magnitude of brain metrics, especially in frontal, temporal, parietal, and occipital lobes under risky situations, while Typically Developing (TD) individuals showed more connectivity in the right hemisphere in the same regions [159]. As for safety cues, higher connectivity is found in the right hemisphere majorly in frontal and occipital regions in ASD, whereas in TD, the changes were found in bi-frontal, left occipital, and in the right parietal regions of the brain [159]. Also, in this study the results support the key concept that ASD individuals have disconnected social cognition network as demonstrated by different graph-theoretic measures, including average weighted degree (AWD), global efficiency, mutual information (MI), centrality and modularity [159].

Modulation. Ewen, et al. [30] has found that in High-Functioning Autism, there is a decreased magnitude of power modulation of EEG during the performance of skilled movements on a praxis task (a motor control task), which is known to be impaired in ASD and has been shown to be associated with the core features of the disorder.

Synchronization. From the connectivity analysis, Han et al. [49] found that several inter-regional synchronizations with reduction were identified in the younger and older ASD groups, and several intra-regional synchronization increases were observed in the older ASD group. Results suggested an age-related decrease of the PL in the delta and theta bands in TD children, whereas atypical age-related alteration was observed in the ASD group. In addition, graph measures were correlated with ASD symptom severity in the alpha band. These results demonstrate that abnormal neural communication is already present at the early stages of brain development in autistic children and this may be involved in the behavioral deficits associated with ASD.

5.1.3 Effective connectivity in Autism

Compared with massive studies of functional brain connectivity, there are few studies which investigated the effective connectivity disorders in Autism subjects. Fortunately, recent researchers are making more efforts and trying to fill this gap.

Granger Causality analysis Granger Causality analysis has been widely used in many studies to investigated the effective connectivity of ASD patients [145]. For example, Tarasi, Luca, et al. [145] has used Granger Causality analysis and found that the directionality of Fronto-Posterior brain connectivity is sensitive to the magnitude of individual autistic traits and the connectivity is mostly conveyed through fast oscillatory activity. Specifically, participants with higher autistic traits were characterized by a prevalence of ascending connections starting from posterior regions ramping the cortical hierarchy [145]. These findings point to the presence of a tendency within the neural mapping of individuals with higher autistic features in conveying proportionally more bottom-up information [145]. Using Granger causality connectivity measures, a recent research has achieved an overall performance of 87.5% accuracy of discriminating a group of 8 autistic individuals and healthy controls [111]. In this research, how communication in a human brain network is affected by autism is also learned by assessing the data from resting

state eight autistic and eight normal subjects along with measures of network connectivity derived from graph theory [111].

Other effective connectivity measures There are many othe measures which can be used to analyze the effective connectivity of the ASD individuals. Talebi, Nasibeh, et al. [143] proposed a causal relationship estimator called nonlinear Causal Relationship Estimation by Artificial Neural Network (nCREANN) that identifies both linear and nonlinear components of effective connectivity in the brain, and the results showed that overall linear connectivity in TD subjects is higher, while the nonlinear connectivity component is more dominant in ASDs. Also, the Temporal Network based on the direct Directed Transfer Function (dDTF) method was used to estimate the effective connectivity. Compared with healthy children, the results showed that the brain connection of ASD children has more diversity and change faster, and it can be also concluded that the information transfer speed of ASD children is less than typically developing children [38]. Moreover, transfer entropy with self-prediction optimality, and four graph theoretic parameters (dergee, clustering coefficient, path length, longest path length) are used to compare the effective connectivity networks of ASD youths with those of healthy controls (HCs) during a passive face processing task [24]. The most remarkable finding in this study is the lower average degree of ASD subjects compared with HC ones is consistent with the under-connectivity theory of ASD, showing the ASD subjects suffer from under-connections which disrupt the information exchange in their brain [24].

5.1.4 Machine learning findings in ASD

In consideration of the atypical patterns in ASD, machine learning has been used to detect children with ASD and has advantages in efficiency and objectiveness compared with standardized diagnostic scales. For example, Liu et al. [83] developed a machine learning algorithm based on face-scanning patterns for classification and identified children with ASD with an accuracy of 82.51%, which indicated promising evidence for applying machine learning algorithms to identify children with ASD. Besides, Jiang and Francis [66] utilized a machine learning method to classify the eye fixations of ASD and TD children, which achieved a classification accuracy of 84%. Besides, during resting-state, the cellular neural network was recognized as an appropriate tool to present the connections and differences between ASD and healthy children patterns in different brain regions, combining with SVM and EEG analysis [6]. In this study, Askari et al. [6] found more abnormalities were related to the connectivity of temporal lobes with other lobes and sults, some significant differences were observed in the states and connections of the frontal lobe and the left hemispheres. This model could distinguish the ASD children from the healthy children with an accuracy rate of 95.1% based on the obtained state and weight values using SVM [6]. In addition, sparse canonical correlation analysis (sCCA) and contrastive principal component analysis (cPCA) were proposed to be able to collectively dissect the ASD heterogeneity in neurophysiology and symptoms, thus further indicating the alteration in functional connectivity by Tong et al. [148]. The results further verified the findings from previous studies, revealing that the right inferior parietal lobe is the core region displaying EEG activity associated with restricted/repetitive behaviors, and functional connectivity between the left angular gyrus and the right middle temporal gyrus is a promising biomarker of social/communication deficits [148].

Deep leaning algorithms, such as Graph Convolution Network (GCN) and Convolutional Neural Network (CNN) are also widely used in the detection and classification of the ASD patients. Zhang et al. [178] first derived multi-channel EEG using the synchronization measures, information entropy, and time-frequency features. Then a random forest was applied to the eye-tracking recordings of the same subjects to single out the most significant features and GCN model then naturally fused the two group of features to differentiate the children with ASD from the TD subjects [178]. This experiment achieved an accuracy of 95% in ASD detection, and illustrated that strong correlations exist between the two bio-signals (EEG and eye-tracking) collected even asynchronously, in particular the EEG synchronization against the face related/joint attentions in terms of covariance [178]. Recently, there was also an increasing interest in the application of Convolutional Neural Networks (CNNs) in ASD detection. Borra et al. [15] proposed a lightweight CNN used in the detection P300 response in EEG data, indicating an average discrimination accuracy improvement from 0.8443 to 0.9227. Moreover, Dong et al. [28] discriminate resting-state EEG with fast reconstructable CNN driven by reinforcement learning. Specifically, the Q-Learning algorithm was utilized to reconstruct the CNN model to reach optimization reusing the tacit knowledge learned from the previous trials [28]. Then a case study has been performed to check

this method versus state-of-the-art counterparts based on resting-state EEG, resulting that the method outperformed the counterparts with an individual/sample accuracy of 92.63%/83.23% [28].

5.2 Depression

5.2.1 Clinical backgrounds

Depression is one of the greatest problems nowadays and is the leading cause of burden worldwide [34]. It is characterized by lack of motivation, difficult to experiment pleasure, impacting on daily lives activities, and in extreme cases, leading to suicide [104]. Symptoms of depressive patients may include a feeling of sadness, hopelessness, and guilty, and other like the loss of interest and concentration, the reduction in energy and changes of appetite and sleep, which may seriously affect their daily life [91]. While the symptoms of the depression can be psychological, the link between the depression and physical symptoms (e.g., headache, constipation, limb pain, stomach problems, pain at joints, back pain, tiredness, appetite and weight changes, sleep changes, and so on) has also been reported [149]. The causes of depression comprise complex relations among psychosomatic, social, and biotic factors and experiences like childhood and teenage adversity, death, major events like losing a job, genetics, and substance abuse may increase the possibility of depression [43]. According to Diagnostic and Statistical Manual of Mental Disorders (DSM-5) of the American Psychiatric Association (APA) [46], depression can exist in various forms, such as Major Depressive Disorder (MDD), Disruptive Mood Dysregulation Disorder (DMDD), Persistent Depressive Disorder (Dysthymia), Premenstrual Dysphoric Disorder (PDD), Substance/Medication-Induced Depressive Disorder (S/M-IDD), Depressive Disorder Due to Another Medical Condition (DDDAMC), and Other Specified Depressive Disorder (OSDD) or Unspecified Depressive Disorder (UDD). Among these, MDD is recognized as the most typical form of depression [46].

5.2.2 Machine learning and deep leaning in Depression

Traditional machine learning-based methods In all detection methods traditional machine learning approaches, it is required to extract relevant features through some non-linear inspection, and then apply a machine learning categorization algorithm like SVM, PNN, Enhanced PNN or ANN. In recent years, deep learning (a sub-field of ANN) has become popular in the medical field for accurate and automatic decision making [146].

CNN based methods CNN is found to be a significant approach in deep learning based depression detection methods. For example, a five layer CNN was trained in classification process using resting EEG signals from 30 normal and 30 depressed persons and attained the highest accuracy of 99.31% from the Right Hemisphere (RH) and 96.3% from the Left Hemisphere (LH) [121]. The results that RH is more prominent than LH for depression patients have also been observed by Seal et al. [123] using a CNN method, DeprNet (consists of five convolutional layers, five batch normalization layers, five max-pooling layers, and three fully connected layers). An accuracy of 99.37% was showed, and on visualizing the last CNN layer, it is found that the values of right electrodes are prominent for depressed subjects, whereas, for normal subjects, the values of left electrodes are prominent [123]. Many researchers have adopted CNN to improve the classification accuracy of depression recognition. However, there are few studies on designing convolution filters for spatial and frequency domain feature learning in different brain regions. Deng et al. [25] has proposed the SparNet, a CNN composed of five parallel convolutional filters and the SENet, to learn EEG space-frequency domain characteristics and distinguish between depressive and normal control. Compared with other methods, SparNet can obtain a relatively good classification performance (with an accuracy of 94.37%). From the results of the local brain regions, it can be seen that the frontal lobe plays a better role in the identification of depression. As for the global brain region, it can be seen that the combination of the spatial features and the frequency domain features can effectively improve the accuracy of depression identification [25]. CNN has much more potential and an increasing number of studies are trying to combine the CNN methods with other techniques, Zhang et al. [179] constructed an 1-D CNN approach for depression detection based on EEG signals by combing temporalspatial convolution with demographic attention mechanism. Medicated and unmedicated MDD patients can be detected using HybridEEGNet, which is composed of two parallel lines and is proposed to learn the synchronous and regional EEG features [160].

CNN-LSTM methods Another robust method is the LSTM, which is the evolution over the RNN to deal with the vanishing gradient problem in the traditional one. LSTM is popular in analyzing time-series data [101] and is widely used in the processing of natural languages, image identification, speech recognition [132]. While CNN has demonstrated promising results in extracting features from stationary data as images, one of the major concerns with CNNs is that it can't assess temporal or earlier data in the time series signals. However, LSTM can be effectively utilized to extract and process the temporal information [150]. So, many studies has proposed that the combination of CNN and LSTM can be utilized to process the EEG signal (Time-Series signal) with better performance. For instance, Sharma et al. [125] has proposed an advanced DepHNN model, which is a Hybrid 6-layer CNN-LSTM based Deep Learning model. The model has less time and minimized computation complexity as it uses the windowing technique and it has attained an accuracy of 99.10% with mean absolute error (MAE) of 0.2040, showing that DepHNN model is accurate, less complex, and useful in detecting depression using EEG signals [125]. Besides, a 12-layer CNN-LSTM has been used automatically detecting depression in terms of the inter-hemisphere imbalance [146]. It has been recognized that depression is often characterised by a hyper-active RH, and a comparatively hypo-active LH. Negative thinking and depression are related to more prominent activities within the frontal area of the RH and the intensity of depression is positively correlated with the hyperactivity of RH [55]. Thus, it is evidence that the accuracy of RH EEG signals is more than LH's during the diagnosis of depression using random-splitting technique (99.07% and 98.84% accuracies for the RH and LH respectively) [146]. Similarly, Ay et al. [8] also used CNN-LSTM network and reached a similar conclusion that right hemisphere EEG will contribute more than the left hemisphere in the recognition process of depression. Using random splitting tests and 10-fold cross validation tests based on EEG signals acquiring from 15 normal and 15 depressed subjects from left half (FP1-T3 channel) and right half (FP2-T4 channel) of the brain, this study was able to detect with 99.12% and 97.66% classification accuracies RH and LH EEG signals, respectively [8]. Deep learning can also be useful for emotion recognition. For example, an advanced 2dCNN+BiLSTM model has been recognized as a novel EEG emotion recognition framework for depression detection. This is a model using short-time Fourier transform and Hilbert–Huang transform to extract time-frequency features, using CNNs to extract spatial features and applying bi-directional LSTM to explore the timing relationship of multi-channel EEG signals, which provides a robust algorithm for real-time clinical depression detection based on EEG. The researchers achieved a recognition rate of over 70% under the 5-fold cross-validation, which exceeded the existing research methods [20]. Moreover, Song et al. [133] proposed an effective end-to-end framework named LSDD-EEGNet and performd the subject-independent experiments across 5 frequency EEG signals and full-bands signals, showing that the gamma-band and full-bands EEG from depressed patients and healthy controls can be well distinguished. Compared to typical machine learning and deep learning models, LSDD-EEGNet outperformed on subject-independent evaluation, which shows the LSDD-EEGNet can be a promising detection method for depression [133].

GCN methods Another popular deep learning method that can be used in depression detection is the graph convolutional neural network (GCN). GCN-based models can naturally exploit the graph structure induced by functional or effective connectivity matrices, and have the potential to capture higher-order topological patterns that may be difficult to encode with conventional convolutional architectures.

Deep learning methods for detection using EEG signals Different deep learning algorithms have been adopted to detecting the depression. Convolutional Neural Network (CNN) is one of deep learning algorithms which can be used to capture the temporal dependencies in the in the time-series EEG input signals, while Long-short Term Memory (LSTM) can be used to process the sequence learning. After the Independent Components Analysis (ICA) and Fast Fourier Transform method (which is used to extract time-frequency information from EEG signals). CNN has the ability to transform EEG input data into cross-sectional data using a technique named windowing. Then, the transformed output is fed into the LSTM block which is in charge of memorizing data from previous steps due to hidden memory cells in its architecture. Different numbers of LSTM and the fully connected layers, which is used to assist the process of automatic detection of depressive cases, are built to construct a model with high performance [117].

Deep learning methods for prediction using EEG signals Deep learning methods combined with the feature extraction can be also used for depression prediction. A research used the LSTM method after removing artifacts of power line interference from recorded EEG signals via a notch filter of 50Hz

and extracting the statistical mean feature via time-domain analysis with moving window segmentation. LSTM block in this study is consist of one LSTM layer with ten hidden neurons, a dropout layer of 0.1 and a dense layer. Comparing with CNN-LSTM and ConvLSTM models, the result shows LSTM model had the best performance with the smallet root mean square error [77]. In order to improving the accuracy of prediction, more advanced techniques can be used and more efforts can be made in the investigation and feature extraction of EEG signals [117].

5.2.3 Clinical outcomes

Clinical outcome studies increasingly link network-level EEG biomarkers of depression to treatment response, relapse risk, and longitudinal symptom trajectories. However, current evidence is still limited by small sample sizes and heterogeneous protocols, and more standardized prospective EEG-follow up studies are needed to clarify how connectivity patterns translate into prognostic tools.

5.3 Epilepsy

Affecting ~ 1 in 100 people worldwide and accounting for 0.6% of the global burden of disease, Epilepsy is the most common chronic brain disease [105]. Epilepsy is characterized by recurrent seizures, which develop with a kind of sudden, recurrent, and transient dysfunction of the central nervous system [14, 131, 181]. Seizures can be controlled using medications and/or invasive surgeries in 70% of diagnosed patients, however, detection or even prediction of seizure onsets remains a challenge due to its high variation patterns of occurrence and manifestation from one individual to another [14]. Thus, a lot of efforts have been given to developing subject-specific, data-driven methodologies, aiming at ultimately improving individualized seizure onset detection and prediction. A number of studies are using the advanced epileptic EEG signals combining with different connectivity measures and/or graph theory to understand the relationship between abnormalities across brain networks of epilepsy patients and the underlying psychophysiology, which will probably improve the current treatment approaches, such as epilepsy surgery, and can potentially improve patient outcomes [14, 103].

Focal epilepsy Focal epilepsy is a dynamic disorder of the brain, which is characterized by both paroxysmal abnormal states (e.g. epileptic seizures), and persistent abnormalities across the functional brain networks. The seizures are disruptions in normal neurological function caused by abnormal, which are often hypersynchronous activity in the brain. The epilepsy can be described as focal when this activity apparently arises from a specific area of the brain [103].

5.3.1 Alternation of functional brain networks

Through various studies, it can be concluded that the variation of the brain function may be because of the damage in functional brain connectivity caused by seizure, which influences the integration and segregation of neural information.

The synchronization of EEG networks Synchronization phenomena of epileptic electroencephalography (EEG) have long been studied. The synchronization patterns (characterized by the algebraic connectivity metric) caused by abnormal neuronal fring during a seizure onset can be detected using a technique for inferring the connectivity of networks (ICON) [71]. Multiple-variate EEG signals has been used in this study to infer a time-varying network, which represents the overall dynamics of the brain network. Based on the observation of 24 patients, the ICON method presented that during a seizure, the strength of interactions between vertices, represented by weighted edges in Figure 5 significantly increases [14]. Overall, this detection method reveals that the brain network properties, such as connectivity and synchronizability, will shift with the occurrence of seizures [14]. Moreover, a sensitivity of 78.79% was achieved, with a false detection rate of 0.02 FPs/h. On average significant changes in the brain activity were observed 1.68 seconds before the actual seizure onset [14]. Apart from this, another detection technique based on the concept of recurrence network (RN), based on which the temporal synchronization patterns are quantified using the spectral graph theoretic features. Using this complex network model to test 23 patients, the results show that the graph theoretic feature yields a high sensitivity ($\sim 98\%$) and low latency (~ 6 s) on average, and seizure onsets in 18 patients are 100% detected. Moreover, the seizure onset can also be characterized by a dramatic increase in synchronization of short time (1 second) temporal proximity between state vectors, which means significant changes in temporal

synchronization epileptic EEG has been found, enabling a patient-specific approach for real-time seizure detection for personalized diagnosis and treatment [32].

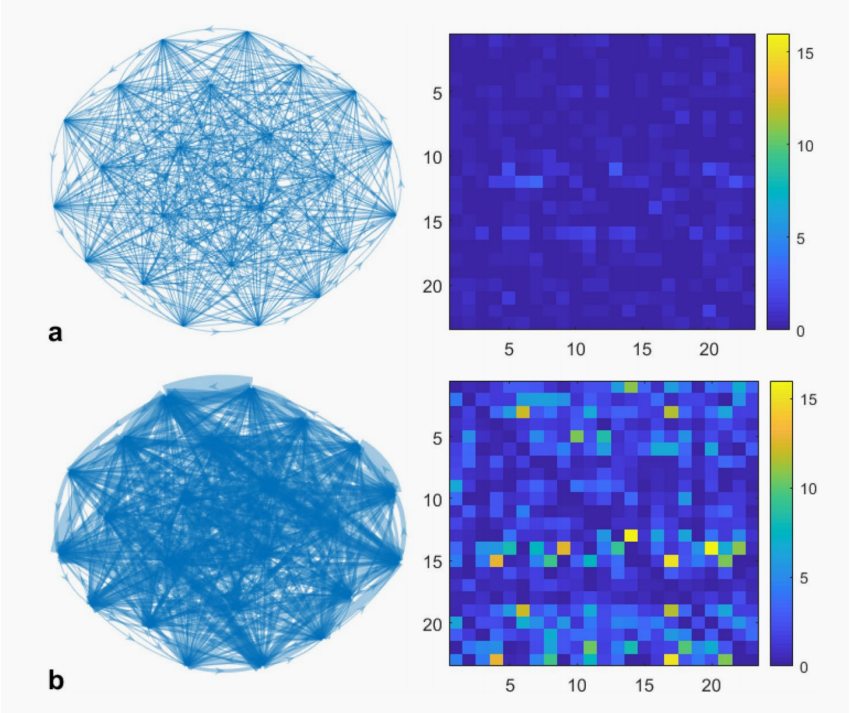


Figure 5: Networks (graph representation and the heat-map representation of the corresponding adjacency matrix) of the normal state (a) and seizure state (b) [14].

Phase synchronization index (PSI) Besides, a recent research applied phase synchronization index (PSI) to investigate the cross-frequency coupling (CFC) of EEG signals and to make comparison with within-frequency coupling (WFC). Reconstructing multi-layered brain networks, including CFC networks and WFC networks and using the graph metrics, this study has investigated the variation of network structure of the epileptic brain. It is observed that, for the CFC, the synchronization level between $\delta - \beta$ and $\delta - \alpha$ couplings are significantly increased during seizure, while that between $\theta - \alpha$ coupling is decreased. For the WFC, the synchronization levels in δ, θ and α bands are increased. Moreover, for WFC networks, the global efficiency of δ, θ and α bands declines, while there is a decline in θ band for local efficiency. For CFC networks, it is observed that global efficiencies of $\delta - \beta$, $\delta - \alpha$ and $\theta - \alpha$ bands show descent trends once during seizure onset, and local efficiencies of $\delta - \beta$, $\delta\alpha$ and $\theta - \alpha$ bands increase simultaneously [173].

Other characteristic analysis of epileptic functional brain network Apart from the ICON method, other epilepsy detection methods have been utilized in other studies, one of them is developed from EEG-based multi-frequency multilayer brain network (MMBN), combining with an attentional mechanism based convolutional neural network (AM-CNN) model [174]. The MMBN analysis has been considered as the lastest development of complex network theory, and this research has shown that MMBN is an excellent detection method with an average accuracy of 99.75% [152]. With the randomly selected eight subjects, four network measures are used, including average clustering coefficient, clustering coefficient entropy, spectral radius and graph energy. The results showed that 93.75% of p-value values are less than 0.001, and 100% of p-value values are less than 0.05. Two randomly selected 8-layer brain networks have been set, where one layer is related to a specific frequency band. The MMBN in the normal state is located in the upper row, and the MMBN in the seizure state is located in the lower row. As can be seen, the network topology shows obvious differences in different frequency bands [152]. Moreover, combining multilayer brain network and AM-CNN model can effectively distinguish between normal state and epileptic state, with a classification accuracy of 99.75% sensitivity of 99.43%, and specifcity of 99.83%. Using the method of complex network and analyzed via graph theory, the resting state EEG signals revealed that for global metrics, there is an increase in the small worldness in focal epilepsy, whereas

for locally, there is a shift towards increased hubness in the left central region and the left parietal lateral surface in the alpha band of focal epilepsy patients [54].

5.3.2 Abnormalities detected using effective connectivity measures

Interictal Connectivity Revealed by Granger Causality Analysiss of sEEG A more advanced technique, Stereoelectroencephalography (sEEG), has facilitated electrical sampling and evaluation of complex deep-seated, dispersed, and multifocal locations. A recent study has found that Granger causality (GC), which was previously used to study seizure networks using interictal data from subdural grids, may assist identifying the seizure-onset zone from interictal sEEG analysis since it can identify seizure foci from interictal sEEG and correlates highly ranked nodes with favorable outcome, thus may potentially informing surgical decision-making without reliance on ictal recordings. This can be shown from a study with 20 patients, using GC, and Wilcoxon rank-sum test. The results reveal that in 12 of 20 cases, the rankings of contacts (based on the sum of outward connection weights) mapped to the seizure-onset zone showed higher causal node connectivity than predicted by chance ($P \leq 0.02$). In 8 of 16, causal connectivity in the resection was significantly greater than in the remaining contacts ($P \leq 0.05$). Additionally, a very low aggregate probability means that causal node connectivity predicts seizure networks and a significant association between favorable outcome and the presence of highly ranked nodes in the resection has been found [139].

Interictal Connectivity Revealed by Granger Causality Analysiss of iEEG Ictal intracranial electroencephalography (iEEG) is used to specify the seizure origin, which is assumed to be the causal part of the seizure network. A previous study examined interictal intracranial electroencephalography (iEEG) recordings from 25 samples using time domain-based conditional Granger causality (GC). After the analysis, the "GC maps" were quantitatively compared to conventionally constructed surgical plans, by using rank order and Cartesian distance statistics. In this study, the set of ictally active electrode (IAE) were used, which in practice, represent the anatomic site of seizure origin. In 16 of 25 cases, the interictal GC rankings of the electrodes in the IAE set were lower than predicted by chance ($P < 0.05$). The aggregate probability of such a match by chance alone is very small ($P < 10^{-20}$) suggesting that interictal GC maps correlated with ictal networks. The distance of the highest GC electrode to the IAE set and to the resection averaged 4 and 6 mm (Wilcoxon $P < 0.001$). Therefore, the results reveal that GC analysis can potentially help localize ictal networks from interictal data, thus predicting the seizure focus [108].

5.3.3 A new approach used the combination of the views of functional and effective connectivity

Interictal disruptions of brain networks in the Bayesian brain Based on predictive coding, free-energy principle and the concept of a hierarchically organized Bayesian brain, a recent study has used a combined view of functional and effective barin connectivity to observed interictal disruptions of brain networks in patients with focal epilepsy [103]. Focal epilepsies are thought to involve widespread functional brain network abnormalities beyond the apparently epileptogenic zone and even during interictal, 'resting state' activity [103]. One particular change at the macroscopic network level is the increased segregation within distributed functional networks [154]. Besides, predictive coding provides a computational framework for understanding how pathophysiological failures of intrinsic connectivity and neuromodulation would lead to false inference and perceptual and motor symptoms. Aberrant prediction-error update in the Bayesian brain would manifest as a blockade on recurrent message passing among different hierarchical levels and a reduction of globally coherent hierarchical neuronal coupling. Persistent reoccurrence of these abnormal changes may result in a loss of global integration between widespread brain functional networks [103].

5.3.4 Detection using complex network techniques

Nowadays, complex network-based epilepsy detection, assisting with the graph-theory are applied in many studies, which can further improve the detection and diagnosis of epilepsy. For example, a study focused on the whole-brain dunamic resting-state functional network revealed that the clustering coefficient is statistically higher in ictal period than in interictal period, and there is no obvious difference for their average path length [50]. As for the degree, a study conducted by L. Wang et al using visibility graphs

to detect the generalized seizures showed that the degree entropy and power-law degree power features increased the performance when combined with other features [163]. During epileptic seizure, assortativity coefficient increased [13] and the mean node degree significantly increased [110]. Moreover, two studies conducted by S. Supriya et al using Weighted Visibility Graph (WVG) to detect the EEG signals during epilepsy seizure and showed high classification performance [140, 141].

5.3.5 Summary: Epilepsy detection based on different methods

Techniques	Author	Year	Database	Features	Results
A technique for inferring the connectivity of networks (ICON)	W. Bomela et al. [14]	2020	CHB-MIT scalp EEG database	Synchronization patterns	The brain network properties, such as connectivity and synchronizability, will shift with the occurrence of seizures.
CFC networks and WFC networks	H. Yu et al. [173]	2020	CHB-MIT scalp EEG database	Phase synchronization index (PSI)	For CFC: The synchronization level between $\delta-\beta$ and $\delta-\alpha$ couplings are significantly increased during seizure, while that between $\theta-\alpha$ coupling is decreased. For WFC: The synchronization levels in δ, θ and α bands are increased.
CFC networks and WFC networks	H. Yu et al. [173]	2020	CHB-MIT scalp EEG database	Local efficiency and Global efficiency	WFC: The global efficiency of δ, θ and α bands declines, while there is a decline in θ band for local efficiency. CFC: Global efficiencies of $\delta-\beta$, $\delta-\alpha$ and $\theta-\alpha$ bands decrease during seizure onset, and local efficiencies of $\delta-\beta$, $\delta\alpha$ and $\theta-\alpha$ bands increase.
GC analysis and sEEG	S. S. D. Stone et al. [139]	2022	Clinical database	Interictal Connectivity	Granger casuality analysis can help identify the seizure foci from interictal sEEG.
GC analysis and iEEG	E. H. Park, J. R. Madsen [108]	2018	Clinical database	Interictal Connectivity	GC analysis can potentially help localize ictal networks from interictal data.

Graph theory and source level resting-state EEG	C. Hatlestad-Hall [54]	2021	Clinical database	Small world index	Global metrics: Increased small worldness in focal epilepsy. Local metrics: a shift towards increased hubness in the left central region and the left parietal lateral surface in the alpha band of focal epilepsy patients.
Graph theory and complex network	C. Hao et al. [50]	2016	Bonn University epileptic database	Clustering coefficient and average path length	Clustering coefficient is higher in ictal period than in interictal period, and there is no obvious difference for their average path length.
Graph theory and complex network	L. Wang et al. [163]	2017	Epilepsy Center Kempenhaeghe	Mean degree, Degree entropy, Power-law degree, coefficient and average shortest path length	Degree entropy and power-law degree power features increased the performance when combined with other features.
Graph theory and complex network	S. Bialonski et al. [13]	2013	Clinical database	Assortativity coefficient	During epileptic seizure, Assortativity coefficient increased.
Graph theory and complex network	G. Petkov et al. [110]	2014	Clinical database	Mean degree, Degree Variance and Local Clustering Coefficient	During epileptic seizure, mean node degree significantly increased.
Graph theory and complex network	S. Supriya et al. [141]	2016	Bonn University epileptic database	Modularity and Average weighted degree	High classification performance.
Graph theory and complex network	S. Supriya et al. [140]	2016	Bonn University epileptic database	Average weighted degree	High classification performance.
EEG-based multi-frequency multi-layer brain network (MMBN)	H. S. Yu and X. F. Meng [174]	2023	CHB-MIT scalp EEG database	Average clustering coefficient, clustering coefficient entropy, spectral radius and graph energy	93.75% of p-value values are less than 0.001, and 100% of p-value values are less than 0.05. The MMBN in the normal state is located in the upper row, and the MMBN in the seizure state is located in the lower row.
EEG-based recurrence network (RN)	M. Fan and C. A. Chou [32]	2018	CHB-MIT scalp EEG database	Temporal Synchronization of EEG signals	A high sensitivity ($\sim$ 98%) and low latency ($\sim$ 6 s) on average, and the temporal synchronization significantly changes in epileptic EEG during seizure onset.

5.3.6 Related clinical outcomes

Network-level EEG markers in focal epilepsy are increasingly explored as potential tools to guide surgical planning, monitor disease progression and quantify treatment effects. However, translating complex

graph-theoretical descriptors into actionable clinical thresholds remains challenging and requires closer collaboration between neuroscientists and epileptologists.

5.4 Schizophrenia (SZ)

Schizophrenia is a serious mental disorder that is characterized by functional impairments, which has been consistently correlated with alterations in cognitive functions. Like bipolar disorder, SZ also has the symptoms like delusions, hallucinations and disorganized discourse [12, 96]. Schizophrenia has been a very promising application of EEG ans different connectivity analysis are able to test the disconnection hypothesis of schizophrenia [35]. Typically, SZ is diagnosed using diagnostic criteria from the Diagnostic and Statistical Manual of Mental Disorders (DSM-5) and asking patients a series of questions, which can elicit information like duration of illness and clinical symptoms [4]. However, the conventional clinical diagnoses may be inaccurate sometimes since the patients may tend to conceal their symptoms and differentiating schizophrenia from other mental illnesses can be tough for experts due to similar symptoms [82]. Therefore, recently, an increasing number of researchers have endeavored to develop more advanced tools for diagnosing patients with SZ for which can be applied to EEG biomarkers.

5.4.1 Functional brain networks analysis using EEG

Resting-state EEG network analysis based on graph theory Like other mental health disorders, graph theory measures has been greatly applied in the study of functional brain networks allows for the description of their general properties and their alterations in Schizophrenia. Among these, many researchers have found a higher global band pre-stimulus Connectivity strength (CS) in the functional brain network of schizophrenia patients [19, 41]. This difference was associated with cognitive deficits and negative symptoms, indicating a hyperactive synchronization among neural groups in schizophrenia that may hamper cognition [19].

Besides, in many studies which applied dynamic network analysis, the functional brain properties of patients with schizophrenia has displayed the smaller clustering coefficient, larger average characteristic path length, lower global efficiency, lower local efficiency and smaller degrees compared with healthy controls [40, 41, 72, 113, 172], indicating there is a stronger fuctional connectivtity in SZ [113]. Based on these findings, Yin, Zhongliang, et al. applied the mutual information (MI) and concluded that there are fewer information interactions in patients than that in healthy controls, while positive SZ has more information interactions than negative SZ does, and the process of transferring information between brain regions patients tend to be slower and less efficient [172]. Kim, Sungkean, et al. [72] has discovered more in detail with graph theory into altered cortical networks using resting-state EEG, which could be promising biomarkers to evaluate patients with schizophrenia. At the global level, patients showed higher strength, CC, and efficiency, and lower PL in the theta band, compared to healthy controls, while at the nodal or regional level, patients showed higher CCs, mostly in the frontal lobe for the theta band [72].

Moreover, in resting-state, mostly decreases (and sometimes increases) in functional connectivity (FC) within and between networks like default-mode network (DMN: for self-related cognitive activity [22]), salience network (SN: for detection of salient stimuli in internal and external environments [22]), and central executive network are reported and can be useful for the study of SZ [100]. For instance, Krukow, Paweł, et al. [76] found that resting-state (baseline) hyperconnectivity the posterior hub of the Default Mode Network, containing posterior cingulate gyrus and precuneus, might impede the ability to initiate cognitive performance in first-episode schizophrenia patients. Another recent study has administered N-methyl-D-aspartate receptor (NMDAR) antagonists to healthy humans, which results in schizophrenia-like symptoms and reached the similar outcome, indicating that there are frequency-dependent increases and decreases in connectivity within and between DMN and SN [22].

Multi-model techniques, such aa simultaneous resting-state functional magnetic resonance imaging (rsfMRI) and EEG recordings has also been applied, and a similar result (Increased FC in the caudate, which may resulting from the disruption of segregation between networks) has been found [73]. However, there are still many conflicts between the FC results due to many reasons. For example, a research calculated for the original EEG data using Phase-Locking Value (PLV), Phase-Lag Index (PLI) and Directed Transfer Function (DTF), and following current source density (CSD) transformation, re-referencing using the average reference electrode (AVERAGE) and reference electrode standardization techniques

(REST) and showed the completely opposite results as the previous studies we reviewed (reduced CS, a lower CC and shorter characteristic path length for both measures of phase synchronization following CSD transformation or REST re-referencing) [102].

Task-related EEG network analysis Spectral entropy (SE) allows comparing task-related modulation of electroencephalogram (EEG) between patients and controls [41]. Using this measure and applying the P300 task, a significant modulation (change between pre-stimulus and response windows) has also been observed in the global and theta bands [40, 41]. Also, Racz, Frigyes Samuel, et al. [113] utilized Entropy analysis to analyze the dynamic functional connectivity (DFC) of resting-state EEG, which showed a reduced temporal complexity of DFC in SZ. Additionally, the facial processing task-related phase synchrony was significantly reduced, and normalized clustering coefficients were significantly increased in schizophrenic patients relative to those of healthy controls [170]. Besides, utilizing the cognitive task, Naim-Feil et al. [95] conducted a study, which indicated patients with schizophrenia exhibited time-sensitive alterations in dynamic patterns of network connectivity across both global and regional measures, reflecting the abnormalities of the cognitive function of patients.

Gamma band activity Abnormal EEG oscillatory activity during schizophrenia, not only in resting-state connectivity analysis, but in event-related connectivity as well [95, 170]. Gamma band activity has been seen to be perturbed in the pathophysiology of Schizophrenia [5, 11, 75, 147, 165]. Changes in gamma activity (30–48 Hz) can lead to disturbed coordination of large-scale brain networks [11]. Yan, Tianyi, et al. [170] investigated into event-related functional connectivity networks and the result revealed that, for the gamma band, there were clear differences in event-related connectivity between patients and healthy controls, especially in the 150–300 ms time window after stimuli onset. It reached the conclusion that there is an increase in local connectivity of brain functional networks during facial processing in schizophrenia [170]. Via resting-state EEG study, an increased power of 20–50Hz in schizophrenia subjects can be observed [165]. This increased gamma activity in certain brain regions that we found may result in disturbances in temporal coordination of resting-state networks in schizophrenia. For example, first-episode Schizophrenia patients displayed an increased mean functional gamma-band connectivity compared to controls in the left rolandic operculum [5]. In the same study, network based analyses indicated increased connectivity within a strongly lateralized network consisting mainly of left inferior frontal/orbitofrontal, lateral and medial temporal, and inferior parietal areas, in which the gamma-band connectivity was higher in patients with low positive and disorganization symptom levels [5]. Moreover, the abnormality in gamma band can be detected using Auditory Steady-State Response (ASSR), and the 40-Hz ASSR may further be recognized as a useful potential biomarker for schizophrenia [75, 147].

5.4.2 Structural brain networks based on EEG

Structural and functional network abnormalities respectively predicted cognitive performance and positive symptoms in patients [40]. In a previous study, patients displayed larger structural PL and pre-stimulus density in the global and theta bands, and lower PL task-related modulation in the theta band [40]. In another study conducted by the same researcher, it is found that structural CLC was associated with SE modulation, which was predictive of negative symptoms, whereas CLC and PL modulation was associated with cognitive performance in the patients [41]. These results support that a hyperactive structural connective deficits has hamper the dynamical modulation of connectivity underlying cognition of the patients [40, 41].

5.4.3 Effective connectivity analysis based on EEG

The studies SZ using effective connectivity measures are rare so far, but there are some researchers making efforts in this domain. For instance, one previous study has utilized Multivariate Transfer Entropy (MTE), which can quantitatively measure the effective connectivity and has been demonstrated to be able to capture both linear and non-linear causal relationships effectively [51]. Based on the analysis using task-related EEG conducted by Harmah, Dennis Joe, et al. [51] and other similar studies, SZ patients most often find it difficult to retain their attention during tasks unlike the HC. Usually the altered brain regions affect the information processing in the SZ and these disruptions give rise to P300 malfunctions, which eventually disturbs the brain at rest in terms of abnormalities [51, 81]. They also compared the results from Granger casuality analysis (GCA) and MTE. Much denser connectivity has been showed for the healthy controls, compared to that of the SZ patients, which extended on the frontal

and parietal lobes. In specific, the corresponding stronger and denser causal connectivity can be found to flow from prefrontal/frontal to parietal lobes. Additionally, compared to the GCA, the MTE gives more causal linkages, shows the dense edges in the frontal lobe [51]. Directed Transfer Function (DTF) has been used by Olejarczyk, Elzbieta, and Wojciech Jernajczyk. [102], indicating the group differences in fronto-posterior asymmetry following CSD transformation and group differences in inter-hemispheric asymmetry for REST.

5.4.4 Machine Learning Classification

Some studies used sensor-level EEG biomarkers, such as event related potential (ERP) amplitude and latency, as features for classification [97, 176]. For example, using machine learning-based classifier, a recent research has been conducted for the differentiation between 49 Schizophrenia patients and 32 healthy controls using features extracted from electroencephalograph (EEG) signals based on ERP. In this study, due to the small dataset, the random forest machine learning algorithm has been chosen to be adopted to different combinations of feature sets for classification, combining with cross-validation using R. The results showed that the accuracy in discriminating healthy control and patients can be improved by adding features extracted from EEG signals [176]. Another similar study sought to identify SZ using amplitudes/latencies of N100 and P300, which were evoked using visual and auditory oddball paradigms respectively, and the result reported a fairly high classification accuracy of 72.4% [97]. In order to avoid some innate limitation of sensor-level features and to improve the accuracy of differentiating the patients and healthy controls, some researchers have utilized both sensor-level features and source-level EEG features for diagnosis [127]. A maximum classification accuracy of 88.24% was reported for the combined-level feature set, which was higher than those of the sensor-level feature set (80.88%) and source-level feature set (85.29%), and the average accuracies were reported to be 78.24%, 72.72%, and 71.40%, for the combined feature set, sensor-level feature set, and source-level feature set respectively [127].

5.4.5 Related clinical outcomes

The effect of drugs As the study conducted by Cea-Canas et al. [19] showed, the global band pre-stimulus CS was significantly higher in the schizophrenia compared with the control groups. Also, theta band CS modulation was decreased in schizophrenia. Besides, these CS values did not significantly alter under treatment with antipsychotics, lithium, benzodiazepines, and anticonvulsants. The first-episode and chronic schizophrenia patients did not show significant differences in CS values. Moreover,Higher global band pre-stimulus CS values were associated with worse general cognition in schizophrenia patients [19]. In addition, de la Salle, Sara, et al. [22] administered a subanesthetic dose of the dissociative anesthetic and NMDAR antagonist, ketamine (which is used to provoke the symptoms of SZ). Ketamine has been found to induce frequency-dependent increases and decreases in connectivity within and between DMN and SN and these altered interactive network couplings together with emergent dissociative symptoms tentatively support an NMDAR-hypofunction hypothesis of disturbed EEG connectivity in SZ [22].

The effect of illness duration The duration of the SZ is strongly associated with long-term treatment with antipsychotic drugs and recent evidence has shown that antipsychotic treatment is associated with loss of cortical gray matter [158]. This has generated a discussion on the effect of "illness duration" on brain networks and functions in SZ. Di Lorenzo, Giorgio, et al. [26] used the Source Functional Connectivity (SFC) Analysis based on resting-state EEG and observed the differences in functional connectivity within and between brain areas of SZ patients with a short duration disease (SDD) and those with a long duration (LDD). In delta and theta bands, SZ patients had greater EEG-SFC than HC and a higher theta band connectivity in frontal regions was observed in LDD compared with SDD. In the alpha band, SCZ showed lower frontal EEG-SFC compared with HC whereas no differences were found between LDD and SDD. Besides, LDD presented lower frontal and parieto-temporal EEG-SFC compared with HC. In the gamma band, SDD had greater connectivity values compared with LDD and HC [26].

5.4.6 Summary: Schizophrenia detection based on EEG networks

Modalities	Techniques	Author	Year	Features	Results
EEG	EEG-based brain networks	Cea-Canas et al. [19]	2020	Phase-locking values	SZ patients experienced an increased connectivity in the whole-brain network, which might hamper cognition in this syndrome.
EEG	Functional EEG network analysis	Gomez-Pilar et al. [42]	2017	Clustering coefficient, characteristic path length and small-worldness	The functional connectivity during cognition in schizophrenia has a larger segregation and a deficit of modulation.
Task-related EEG	Graph theory and dynamic network analysis	Naim-Feil et al. [95]	2018	Degree, Strength, Clustering Coefficient, Path Length and Global Efficiency	Patients with SZ exhibited time-sensitive alterations in dynamic patterns of network connectivity across both global and regional measures, reflecting abnormalities in the flexibility of the network that underlies aspects of cognitive function.
Resting-state EEG	Graph theory based source-level weighted functional networks	Kim, Sungkean, et al. [72]	2020	Strength, Clustering coefficient, Path length, and Efficiency	At the global level, SZ patients showed higher strength, CC, and efficiency, and lower PL in the theta band, while at the nodal level, they showed higher CCs mostly in the frontal lobe for the theta band. Alterations in cortical networks were correlated with core symptoms of SZ.
Resting-state EEG	EEG Source Functional Connectivity Analysis	Di Lorenzo, Giorgio, et al. [26]	2015	Lagged phase synchronization	Resting state brain network connectivity is abnormally organized in SZ and may be influced by the illness duration.
Task-related EEG	Graph theory and facial processing	Yan, Tianyi, et al. [170]	2017	Phase lag index	Task-related phase synchrony was significantly reduced distributed networks, and normalized clustering coefficients were significantly increased in schizophrenic patients relative to healthy controls.

Task-related EEG	Graph theory and facial processing	Yan, Tianyi, et al. [170]	2017	Phase lag index	During facial processing, phase synchrony of the networks was significantly reduced and the local connectivity of brain functional networks was increased in SZ patients relative to healthy controls.
Task-related EEG	Graph theory and odd-ball task	Yan, Tianyi, et al. [170]	2017	Averaged path-length, clustering coefficient, and density	During odd-ball task, phase synchrony of the networks was significantly reduced and the local connectivity of brain functional networks was increased in SZ patients relative to healthy controls.
Task-related EEG	EEG effective network, Granger causal analysis, Multivariate transfer entropy and P-300 task	Harmah, Dennis Joe, et al. [51]	2020	Topological Differences	During P-300 task, much denser connectivity has been showed for the HCs, compared to that of the SZ patients, which extended on the frontal and parietal lobes and MTE outperformed GCA in this task.
EEG	Functional brain networks based on Mutual information (MI) and Graph theory	Yin, Zhongliang, et al. [172]	2017	Path length, average clustering coefficient, efficiency and degree	There are fewer information interactions in SZ patients than that in HCs, while positive SZ has more information interactions than negative SZ does, and patients tend to have slower and less efficient information transfer between brain regions.
Task-related EEG	P-300 task, functional networks and structural networks	Gomez-Pilar, Javier, et al. [41]	2018	Averaged clustering coefficient, characteristic path-length and connectivity strength	A hyperactive functional connectivity and/or structural connective deficits in the patients hamper the dynamical modulation of connectivity underlying cognition.

EEG	Dynamic functional connectivity, multi-fractal analysis and information-theoretical entropy	Racz, Frigyes Samuel, et al. [113]	2020	Connectivity strength, Clustering coefficient and Global efficiency	There is an increased FC and a reduced temporal complexity of DFC in SZ.
rsfMRI-EEG	CONN toolbox	Kirino, Eiji, et al. [73]	2019	Coherence	There is an increased FC in the caudate, representing aberrant between-network FC resulting from the disruption of segregation between networks.
Resting-state EEG	Functional connectivity	Andreou, Christina, et al. [5]	2015	Power envelope correlation	The between and within networks functional gamma-band connectivity increased in SZ patients, which is associated with the core symptoms of SZ.
Resting-state EEG	Functional connectivity	Baradits, Máté, et al. [11]	2019	Power measures	Positive association of increased gamma power with psychopathology suggests that altered gamma activity may result in disturbances in temporal coordination of resting-state networks in SZ, providing a contribution to symptom presentation.
Resting-state EEG	Dynamic analysis and Default Mode Network(DMN)	Krukow, Paweł, et al. [76]	2020	The non-parametric randomization approach (used to identify hypo- and hyper-connections) and Synchronization	Baseline hyperconnectivity within the posterior hub of the DMN, containing posterior cingulate gyrus and precuneus, might disturb effective cognitive outcome.

5.5 Anxiety disorder

Affecting ∼278 patients worldwide, anxiety disorder arises from dysfunctions in neural signaling and transmission pathways of the brain [124]. Behavior exhibited by patients with generalized anxiety disorder (GAD), a subtype of anxiety disorder, include persisting feelings of nervousness or stress that are often associated with the patient's life problems. Diagnosis criteria and treatment consider arousal, the state of awareness to external stimuli and over-expression of which is linked to anxiety disorder.

Machine Learning Classification Multidimensional EEG feature extraction used alongside machine learning may be an accurate mechanism for automatic GAD detection. This study aims to detect GAD by its effect on functional connectivity (FC), the interconnectedness of different brain regions and the correlation between regional activity, providing a quantitative approach to identify the presence of GAD possibly useful for diagnostic criteria. Forty-five patients and thirty-six healthy controls (HC) were measured with a 16 electrode EEG system, and the resulting signals were analyzed multidimensionally using three conventional extraction techniques: power-spectral-density (PSD) analysis, fuzzy entropy

(FE) analysis, and FC analysis. The study yielded an exceptional GAD classification accuracy of up to $97.83 \pm 0.40\%$, a notable improvement from previous studies [126].

EEG-based Neurofeedback Another study examines the use of EEG-based neurofeedback (EEG NF) for the treatment of anxiety disorder. EEG measures and visually displays the patient's neural oscillations, providing self-regulated and non-invasive treatment. The patient and technology engage in a closed-loop process as the EEG pattern classified by machine learning algorithms rehabilitates the patient through continuous feedback [90]. Neurofeedback (NF) for over-arousal generally has three approaches: up-regulation of low-frequency rhythms (e.g. theta or alpha), down-regulation of high-frequency rhythms (e.g. beta and gamma), or reduction of low-frequency rhythms to stimulate post-NF rebound and normalization [90]. Alpha rhythm (8–12Hz) is a known biomarker of arousal/cortical excitation in reduced levels and relaxation in heightened levels [23]. Hence, one approach for recent studies is to utilize EEG detection of alpha power signaling the current state of stress/arousal to facilitate self-regulated NF. In one clinical study, forty-five subjects with generalized anxiety randomly received either one of four biofeedback treatments regulating alpha or the control condition, and decreased anxiety was observed in all treatments with the most obvious signs of reduced anxiety in the alpha-increase biofeedback subjects [115].

5.6 Alzheimer's disease (AD)

Alzheimer's disease (AD) has become the most common neurodegenerative disorder characterized by cognitive and intellectual deficits and behavior disturbance [65]. In the pre-clinical stage of AD, no reliable or valid symptoms can be detected to allow a very early diagnosis before the manifestation of irreversible cognitive deficits. In the mild stage, the impairment of learning and memory, semantic difficulties with word generation and a deterioration of object naming are often prominent. Language difficulties become more obvious as word finding difficulties, paraphasia, and the circumstanciality increase in the moderate dementia stage, as well as other cognitive abilities, such as judgment, abstract or logical reasoning, planning, and organizing, may appear during the progression of the disease. In the late stage, almost all cognitive functions are severely damaged, with motor functions including chewing and swallowing are profoundly disturbed [65].

Resting-state EEG Visual EEG analysis links AD to a prominent decrease in alpha power and increase in delta, as well as abnormalities in spectral coherence, the synchronization of oscillations in different neuronal populations [57]. Studies that used a quantitative approach, computerized EEG spectral analysis, have reinforced the correlation of decreased mean frequency of oscillations in AD patients as opposed to those of controls [65].

Connectivity analysis Isolated neural responses are not always accurate indicators of the presence of AD, so many recent studies utilizing EEG have focused instead on functional connectivity (FC). The benefit of utilizing an FC measure that is nonlinear rather than linear is the potential of higher sensitivity and lower bias particularly concerning volume conduction and low spatial resolution. One such nonlinear method recently developed is "lagged phase synchronization", using the Exact Low Resolution Electromagnetic Tomography (eLORETA) statistical package [53, 109]. Lagged phase synchronization is considered to be a correction compared to other measures of FC because it is able to evaluate only signals of physiological connectivity, without the biased influence of non-physiological components or physical artifacts. One study utilizing this method showed that patients with AD displayed a significant decrease in types of lagged phase synchronization in most cortical regions and between the right dorsolateral prefrontal cortex (DLPFC) and right posterior inferior parietal lobule (pIPL), compared to healthy controls [53].

5.7 Other Mental Health Disorders

5.7.1 Obsessive-compulsive disorder (OCD)

Obsessive–compulsive disorder (OCD) is a common neuropsychiatric disorder characterized by recurrent intrusive thoughts (obsessions) and repetitive behaviours (compulsions) with a prevalence of 2˘3% [69]. EEG-based studies, including connectivity and spectral analyses, have started to uncover alterations in cortico-striato-thalamo-cortical circuits, but the literature is still relatively sparse compared with other disorders reviewed above.

5.7.2 Post-traumatic stress disorder (PTSD)

Post-traumatic stress disorder (PTSD) patients suffer from various symptoms, such as re-experience, avoidance, and hyperarousal [128]. This usually occurs when a person's ability to function is hampered by emotional responses to recollections of terrible experiences and those patients frequently have intrusive, unwanted and upsetting thoughts, which may be followed by frequent nightmares, flashbacks, and intensely negative responses to inputs like sounds and touch [85]. Due to a high rate of comorbidity with other psychiatric disorders (such as depressive and anxiety disorder), there is a high probability of misdiagnosis of PTSD [29]. EEG connectivity and graph-theoretical studies in PTSD remain relatively limited, but emerging work suggests alterations in networks supporting threat detection, salience processing, and emotion regulation.

6 Discussion

In this section, we synthesize the findings across different mental health disorders and discuss common themes, methodological considerations, and remaining challenges for clinical translation of EEG-based network analyses and machine learning models.

6.1 Common network-level patterns across disorders

Across ASD, depression, epilepsy, schizophrenia, anxiety disorders, and other conditions, a recurring theme is that mental health disorders are associated with systematic alterations in brain network organization rather than isolated abnormalities at single electrodes. Many studies report deviations from "small-world" topology, including changes in clustering coefficient, path length, global and local efficiency, and hub structure. For example, ASD and schizophrenia often present with disrupted long-range connectivity and altered local clustering, whereas epilepsy is characterized by dynamic shifts in synchronization and network integration around seizure events. In depression and anxiety, hemispheric asymmetries and changes in frontal networks are prominent, while AD and other neurodegenerative disorders show progressive loss of connectivity in higher-order association networks.

Despite disorder-specific patterns, some broad motifs appear repeatedly: (i) abnormal balance between segregation and integration (either excessive segregation or pathologically increased global synchronization), (ii) frequency-band specific alterations (e.g., theta and alpha in ASD and SZ, gamma-band changes in SZ, low-frequency increases in AD), and (iii) developmental or illness-stage effects, where the direction and magnitude of connectivity changes can flip across childhood, adolescence, and chronic stages. These converging observations support the view that many mental disorders can be conceptualized as "dysconnection syndromes" at the network level, even though the precise manifestation and clinical correlates differ by diagnosis.

6.2 Methodological considerations: functional vs. effective connectivity

The reviewed work also highlights the methodological diversity in how EEG connectivity is defined and quantified. Functional connectivity measures such as coherence, phase locking value, power envelope correlations, and lagged phase synchronization are extensively used to capture statistical dependencies between signals. Graph theory provides a unifying framework to transform these pairwise measures into network-level descriptors and to compare patients with controls in terms of small-worldness, hubness, and modular structure.

Effective connectivity methods, including Granger causality, directed transfer function, and multivariate transfer entropy, go one step further by probing the directionality of information flow. This is particularly relevant in epilepsy, where identifying seizure onset zones and propagation pathways can guide surgical decisions, and in ASD and SZ, where altered bottom-up vs. top-down signaling has been hypothesized. However, effective connectivity analyses are more sensitive to preprocessing choices, model assumptions, and data quality, and cross-study comparisons remain difficult. Taken together, functional and effective connectivity should be viewed as complementary perspectives; combining them, for example within Bayesian brain or predictive coding frameworks, may yield more mechanistic insight than either alone.

6.3 Machine learning and deep learning on EEG networks

Traditional machine learning approaches (e.g., SVM, random forest) using hand-crafted spectral, temporal, and network features have achieved promising classification accuracies across disorders, including ASD, depression, epilepsy, SZ, and anxiety. More recently, deep learning models such as CNNs, CNN–LSTMs, GCNs, and hybrid architectures that integrate EEG with other modalities (e.g., eye-tracking) have further improved performance, in some cases reaching accuracy levels above 90% in carefully controlled datasets.

At the same time, several caveats are evident. Many studies rely on relatively small samples or single-center data, which limits generalizability. Cross-validation procedures are not always strictly subject-independent, raising the risk of optimistic performance estimates. Deep models often lack interpretability, making it difficult to link decision rules back to pathophysiological mechanisms or to ensure robustness across recording conditions. Future work will need to combine the representational power of deep learning with principled feature engineering, explicit modeling of connectivity structure (e.g., via GCNs on functional or effective networks), and rigorous evaluation protocols that emphasize subject-level generalization and cross-dataset robustness.

6.4 Clinical translation and open challenges

From a clinical perspective, EEG-based network and machine learning methods hold several attractive properties: they are relatively low-cost, non-invasive, and suitable for repeated measurement across development or disease course. Potential applications include early detection (e.g., ASD, depression, AD), individualized seizure prediction and localization (epilepsy), stratification of heterogeneous syndromes (ASD, SZ, PTSD), and treatment monitoring (pharmacotherapy, neuromodulation, and neurofeedback).

However, several gaps must be addressed before these methods can be routinely adopted in practice. First, there is a need for larger, multi-center datasets with harmonized acquisition protocols and rich clinical annotations to enable robust model training and external validation. Second, methodological choices in connectivity estimation, graph construction, and model training should be standardized or at least systematically compared to avoid conflicting results driven by analysis pipelines. Third, translating high-dimensional network descriptors into simple, clinically interpretable markers (e.g., risk scores, staging indicators, or treatment response biomarkers) requires new work at the interface of statistics, machine learning, and clinical decision-making. Finally, ethical and fairness considerations are essential, as EEG-based models may perform differently across age groups, sexes, and cultural contexts if training data are not representative.

Overall, the current literature shows that EEG-based functional and effective connectivity analysis, combined with advanced machine learning, can capture meaningful signatures of a wide range of mental health disorders. The next step is to move from proof-of-concept classification studies towards robust, interpretable, and clinically actionable tools that can support diagnosis, prognosis, and personalized intervention in real-world settings.

7 Summary

In summary, this review has outlined how EEG-based functional and effective connectivity, together with graph theory and machine learning, can illuminate the network-level alterations underpinning autism, depression, epilepsy, schizophrenia, anxiety, AD, and other mental health conditions. While each disorder shows distinct patterns in terms of affected networks, frequency bands, and developmental trajectories, common themes emerge around disrupted integration–segregation balance and altered information flow. Deep learning models further demonstrate that these patterns can be leveraged for accurate automated detection and prediction, yet challenges related to data scale, methodological heterogeneity, interpretability, and clinical translation remain. Addressing these challenges will be crucial for turning EEG network biomarkers from promising research tools into reliable components of future precision psychiatry and neurology.

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